

Working Group C32

Protection Challenges and Practices for Interconnecting Inverter Based Resources to Utility Transmission Systems

**Power System Relaying and Control Committee
Report of Working Group C32
of the
System Protection Subcommittee**

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Synchronous Generator
Synchronous Condenser
Static Synchronous Condenser
Sub Synchronous Control Interaction
Sub Synchronous Oscillations
Directional Relay
Distance Relay
Power Swing
Auto-Reclose
Single-Phase Trip and Reclose

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1. INTRODUCTION

Rapid growth in interconnection of solar photovoltaic and Type III or IV wind energy conversion sources to the transmission system is creating new challenges for protection engineers. Figure 1 and Figure 2 illustrate basic configurations of solar and Type IV wind conversion resources, respectively. They have a full power electronic converter interface between the grid and the resource. The interface is sized based on the total power output of the generation. Type III wind energy conversion resource is a doubly fed asynchronous generator whose stator directly connects to the grid and the rotor through a power electronic converter as shown in Figure 3. In this case, the interface is sized based on a fraction (about 30%) of the total generation capacity. All three types of resources are referred to as inverter-based resource (IBR) for convenience in this report.

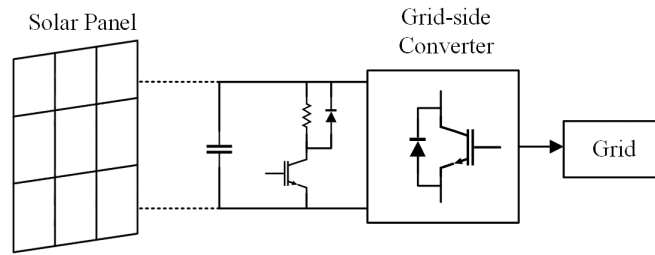


Figure 1: Basic configuration of solar photovoltaic (PV) energy conversion source

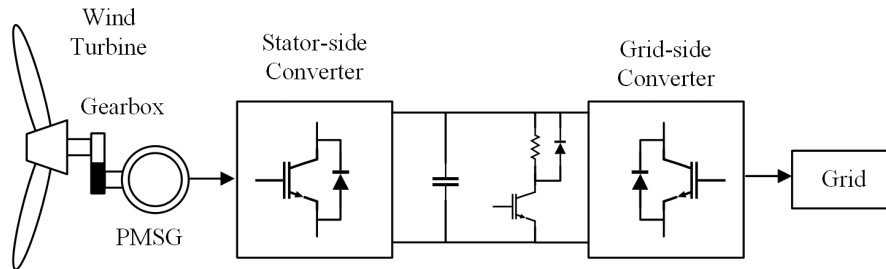


Figure 2: Basic configuration of Type IV wind energy conversion source

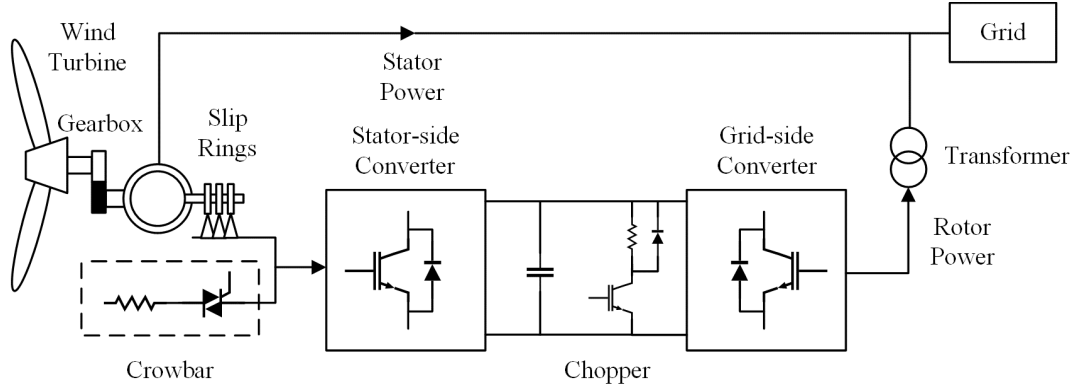


Figure 3: Basic configuration of Type III wind energy conversion source

Protection challenges are introduced because the output current of an IBR facility is very different from a traditional rotating synchronous source facility during short circuit conditions. Current from a synchronous source, immediately after a short circuit and within the timeframe of protection operation, is of high magnitude, uncontrolled and can be mostly defined by electrical parameters of the source and impedance of short circuit path. However, the short circuit current of the IBR is of low magnitude, highly controlled and managed by using fast switching of power electronics devices dependent upon manufacturer specific and often proprietary control system design. One of the key control design objectives is limiting the magnitude of current within the thermal withstand capability of power electronics during short circuits while meeting the grid code requirements (if applicable). Thus, an IBR has low-magnitude and non-universal short circuit current characteristic.

Traditional protection schemes, which largely rely upon high magnitude and high inductive nature of the short circuit current, may not provide reliable protection when operating on controlled current supplied by an IBR. Not taking into account the differing nature of IBR behavior, traditional line protection may incorrectly trip for some external short circuits or may not trip on other internal short circuits. In addition, weather conditions introduce large changes in IBR output and that may have a negative influence on line protection reliability. This report describes protection challenges associated with interconnection of IBR facilities, suggests solutions, and documents lessons learned from the present limited experiences thus far. The emphasis is transmission or sub-transmission system protection issues on the point of interconnection (POI). This report does not address specific issues associated with interconnection of IBR to utility or privately-owned distribution systems or microgrids. The goal of the report is to provide a resource to assist protection engineers in the successful integration of IBR to the electric power grid using multifunction relaying devices available in the marketplace.

1.1 Purpose

IEEE C37.246-2017 Guide for Protection Systems of Transmission to Generation Interconnections [1] describes integration of generation facilities into the transmission network. It describes various interconnection configurations, their advantages and disadvantages, protection problems and their solutions to address them. This report focuses only on protection issues posed by transmission interconnection of IBR over and above those already discussed in the published guide [1].

Figure 4 shows transmission interconnection of two inverter-based generating stations to the integrated power system. The solar generating station is interconnected to the grid through a line that already has a tapped transmission customer, whereas the wind turbine generating station is interconnected through a dedicated line. Two conventional generating stations (CG1 and CG2) within the integrated power system are comprised of synchronous sources whose size and short circuit strength are significantly more than either of the inverter-based generating stations. This figure is referred back in various sections of this report when highlighting issues related to traditional protection systems operating on the short current supplied solely or significantly by inverter-based generator at locations such as at CB1 (Circuit Breaker 1) or CB2 (Circuit Breaker 2).

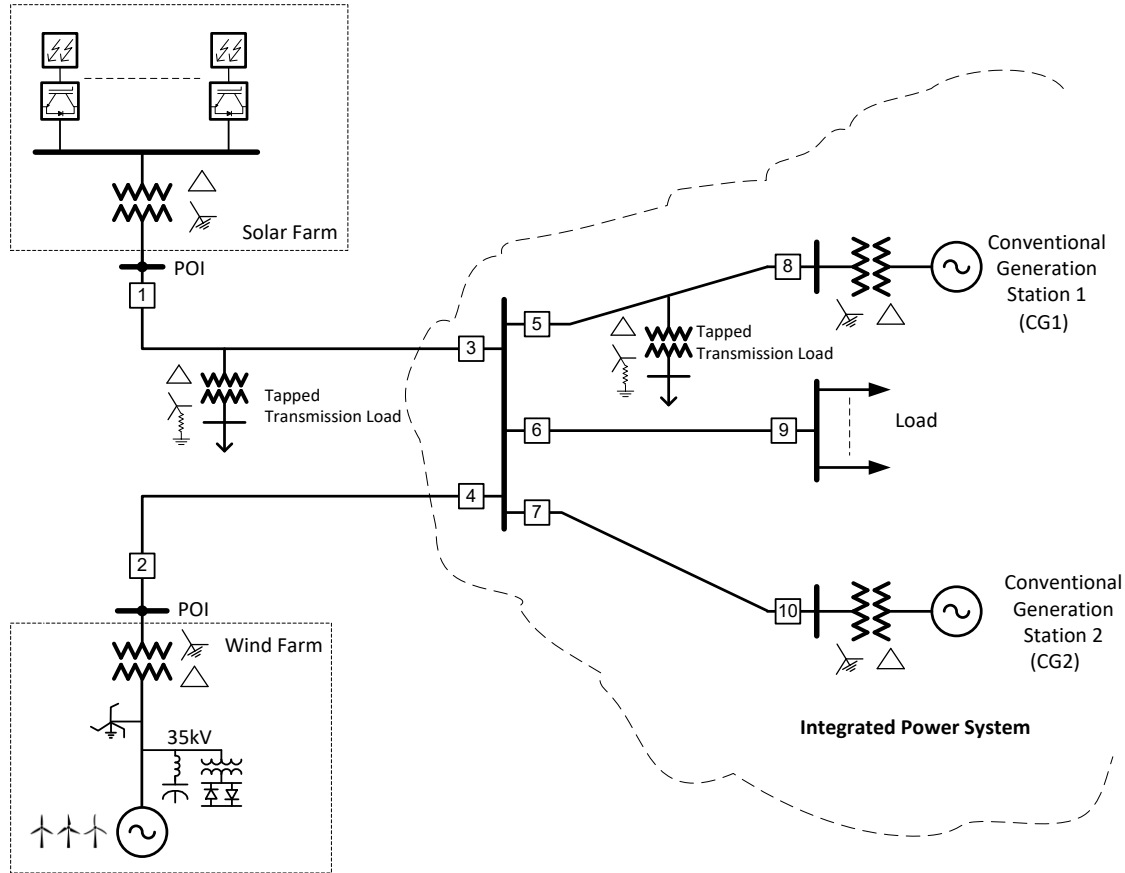


Figure 4: Sample interconnection configurations

The broad purpose of this report, new challenges to the traditional line protection schemes from interconnecting IBR facilities, are highlighted with the help of recorded real-world fault data. The following example was an internal phase-to-phase fault (A-to-C) on a short 138 kV line interconnecting a 100 MW Type IV wind turbine facility to the grid. The interconnection configuration was similar to Figure 4 where the wind turbine facility was connected by the line between CB2 (Circuit Breaker 2) and CB4 (Circuit Breaker 4). Figure 5 shows the short circuit current waveforms captured by the line protection relays. These waveforms were filtered by a 60 Hz bandpass filter within the line relay design and were stored at a rate of 4 samples per cycle. The upper plot is three-phase current from the relay at CB4. This line relay was set to look into the line and operate on short circuit currents supplied by the integrated system to the fault. Likewise, the lower plot is recorded data from the relay located at CB2 operating on short circuit current supplied by the wind turbine facility.

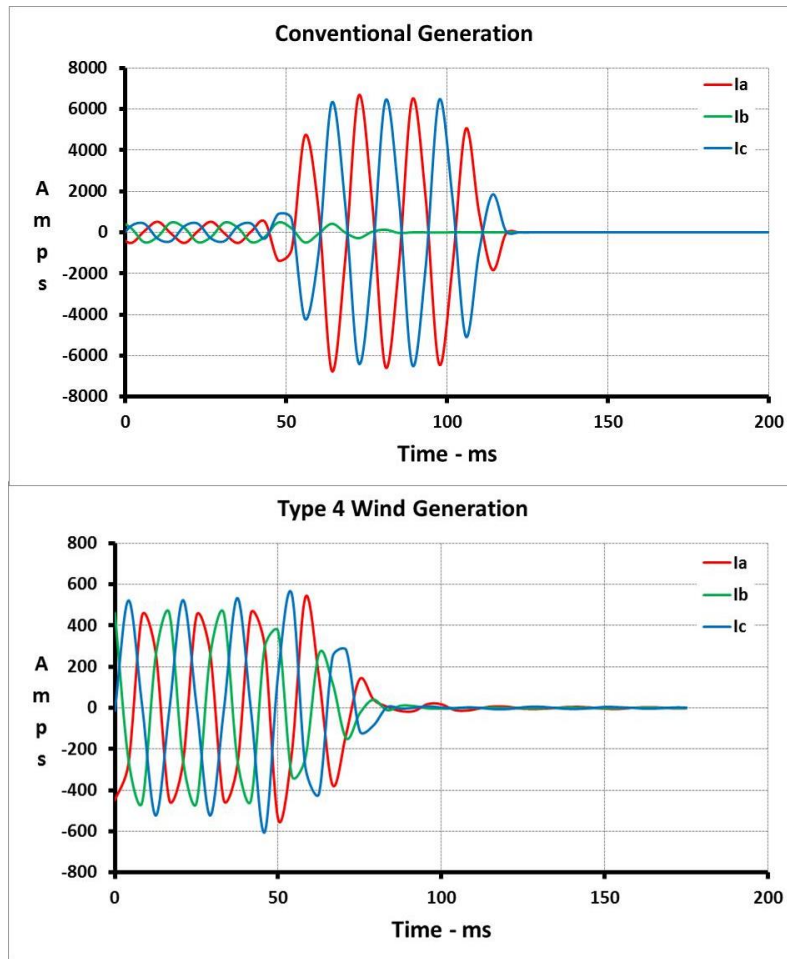


Figure 5: Dissimilar short circuit responses of conventional generation and Type IV wind turbine facilities to a phase-to-phase fault

Prior to the fault, the wind facility was exporting rated power output at unity power factor. The short circuit current from the integrated system, comprised of many interconnected synchronous generators, was as expected. The currents in the two faulted phases increased

significantly from the pre-fault 420 A to more than 6000 A during the fault. The line distance relay (Zone 1) correctly operated and tripped CB4 within three cycles. However, the controlled short circuit response of the wind turbine facility was significantly different. The short circuit currents from the wind turbine facility did not increase at all on fault inception and thereafter. It remained at its pre-fault level and then started to rapidly drop after about two cycles into the fault becoming almost zero around the end of the third cycle. In this case, the IBR could be essentially viewed as a source whose internal impedance rapidly increased and became an open circuit in about three cycles after the fault inception. The protection application engineer was not familiar with the specific control system used in the facility and did not anticipate the short circuit response as observed. Thus the traditional line distance protection scheme applied at CB2 did not respond to this phase-to-phase in-zone fault.

The example presented illustrates the significantly different short circuit response of an IBR compared to the integrated system or cluster of interconnected conventional synchronous sources. It highlights that the reliability of traditional line protection schemes, operating solely or largely on short circuit current contributions from IBR facilities can be unreliable without additional measures. Line protection systems, located at the integrated bus and operating on high short circuit strength from many interconnected conventional generators, can also be at risk during system contingencies. As an example, the protection system at CB5 (Circuit Breaker 5) in Figure 4 can be at risk when it is expected to operate on short circuit current from the IBR facility under contingency of CG2 being out-of-service or likewise at CB7 (Circuit Breaker 7) when CG1 is out-of-service. This report uses several examples of the actual recorded faults, supplemented by a few simulated faults, to illustrate line protection reliability challenges posed by a high penetration of IBR and discusses countermeasures that can either mitigate or minimize their reliability risk.

1.2 Grid Code

As the size of IBR facilities started to increase and their installed capacity within a transmission system began to rise, transmission planners started to recognize system integration challenges. Utilities and the regulators around the world in-turn introduced grid codes with additional requirements to connect the IBR facilities. These interconnection requirements influenced control system designs of the IBRs, thereby their short circuit current outputs and consequently protection responses. Most utilities or utility regulators now have their own interconnection requirements. Using the German grid code as an example, this section introduces and illustrates the relevance of the code to the line protection systems with IBR facilities. Finally, this section also suggests the introduction of a new grid code, similar to the Germany grid code, for the benefit of protection reliability during unbalanced faults.

1.2.1 IBR Low Voltage Ride Through

Figure 6 shows low voltage ride through requirement imposed by the German grid code [2] for the IBR wanting to connect to its transmission system. Most utilities and reliability regulators have similar low voltage ride through requirements, with some variation. According to this requirement, the source is to remain connected when the voltage depression from an external fault is within the low voltage ride through requirement of the

applicable grid code. Because of this requirement, the short circuit current characteristics of the IBRs have become relevant to the line protection security. An undesirable line protection operation during an external short circuit will disconnect the generator and defeat requirements specified by the grid code.

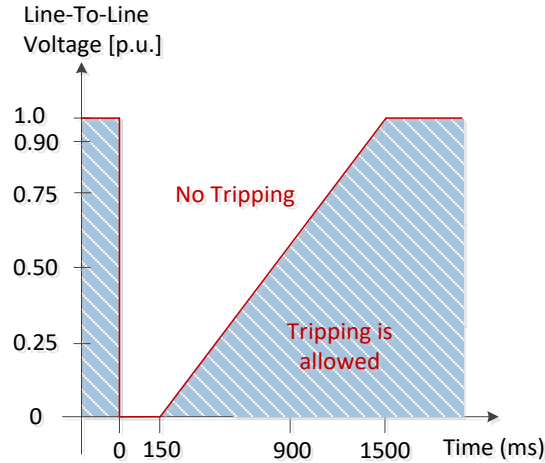


Figure 6: Low voltage ride through requirement by the German grid code

1.2.2 IBR Positive Sequence Reactive Current Injection

Figure 7 illustrates dynamic voltage support of the German grid code [2] during short term voltage drops from IBR wanting to connect to its transmission grid. In case of voltage depression, the source is required to supply positive sequence reactive current, irrespective of fault type. This increased reactive current is required to minimize the voltage depression during faults and assist in preventing loads such as induction motors from stalling and further depressing the voltage. Similarly, the source is required to absorb positive sequence reactive current during overvoltage conditions, to thereby help in returning the voltage to a normal level.

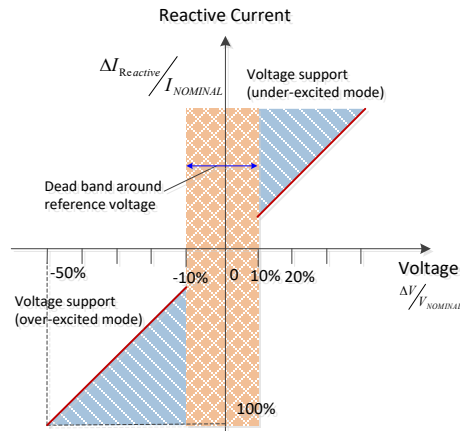


Figure 7: Positive sequence dynamic reactive current requirements during voltage excursions

1.2.3 IBR Negative Sequence Reactive Current Injection

Negative sequence relaying became easily available after the introduction of multifunction microprocessor-based relays in the marketplace. Because of benefits such as negative sequence elements being unaffected by zero sequence mutual coupling and improved sensitivity to high resistance ground faults [3], utilities started to widely apply negative sequence relaying in transmission line protection applications for unbalanced faults, particularly for the ground faults. Except for one recently introduced new grid code VDE-AR-N 4130 in Germany [4], none of the other utilities (at the time of this writing) has a requirement concerning negative sequence current injection. Hence, the IBR manufacturers typically in their present day control system designs suppress negative sequence current. Absence of adequate negative sequence current during unbalanced faults, as a consequence, is threatening protection reliability i.e. its ability to adequately protect the line during unbalanced faults.

This report will illustrate that universal form of negative sequence reactive current injection requirement proportional to negative sequence voltage unbalance, as shown in Figure 8, will aid reliability of protection system operation on short circuit currents from IBR. The k is the line slope of the negative sequence reactive current injection requirement. An IBR with adjustable line slope i.e. gain (k) from 2 to 6 may emulate apparent $j1/k$ pu negative sequence reactance largely similar to a synchronous generator which typically has negative sequence reactance in range from $j0.12$ pu to $j0.4$ pu. However, the final amount of injected negative sequence reactive current depends also on the current limiter of the inverter. Also, the IBR control system takes time to measure the voltage unbalance and respond by injecting the negative sequence reactive current and settle down to the required level. This time delay is one or two cycles or up to the maximum delay permitted by the grid code. The IBR transient response during this time can negatively impact the reliability of the high-speed line protection system.

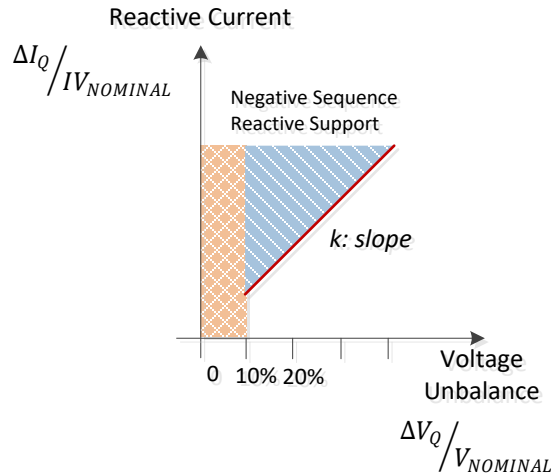


Figure 8: Proposed requirement for negative sequence reactive current injection proportional to negative voltage unbalance

It should be noted that IBRs are made of power-electronic current controlled devices. Even when required, injection of a negative sequence current during unbalanced faults from inverter-based resources would be limited by the inverter ratings and the required positive sequence reactive injection. As an example, assume a 100 MVA conventional synchronous generator having 15% sub-transient reactance is interconnected to the system through a transformer having 10% leakage impedance. A line-to-line transmission line fault close to the transformer may result in about 50% negative sequence voltage and about 200% negative sequence current, irrespective of load current. In contrast, assuming a 100 MVA IBR having its current limit capped at 120% would only be able to inject for a similar fault approximately 20% negative sequence current while maintaining pre-fault 100% load current. Note that depending on type and location of a fault, flow of pre-fault load current between the IBR and loads may not be possible. As an example, for a bolted three-phase fault between IBR and load, the load current cannot flow from IBR to load because of zero voltage between two. However, maintaining pre-fault load current injection during a fault allows for determination of lowest possible negative sequence current injection during unbalanced faults. Figure 9 shows estimates of the current through an inverter that supplies negative sequence current for a close-in line-to-line fault on a transmission line. It illustrates that current in at least one phase of the inverter hits a limit of 120%, which in turn limits the negative sequence current injection during a fault. The figure illustrates that the amount of negative sequence current injection would increase to about 60%, if the IBR is not required to inject pre-fault load current during a fault. The intent of this example is to illustrate that even when negative sequence current injection is required, the magnitude would be limited because the vector sum of load current (if present), positive sequence reactive current and negative sequence reactive current cannot exceed the current limit of an IBR.

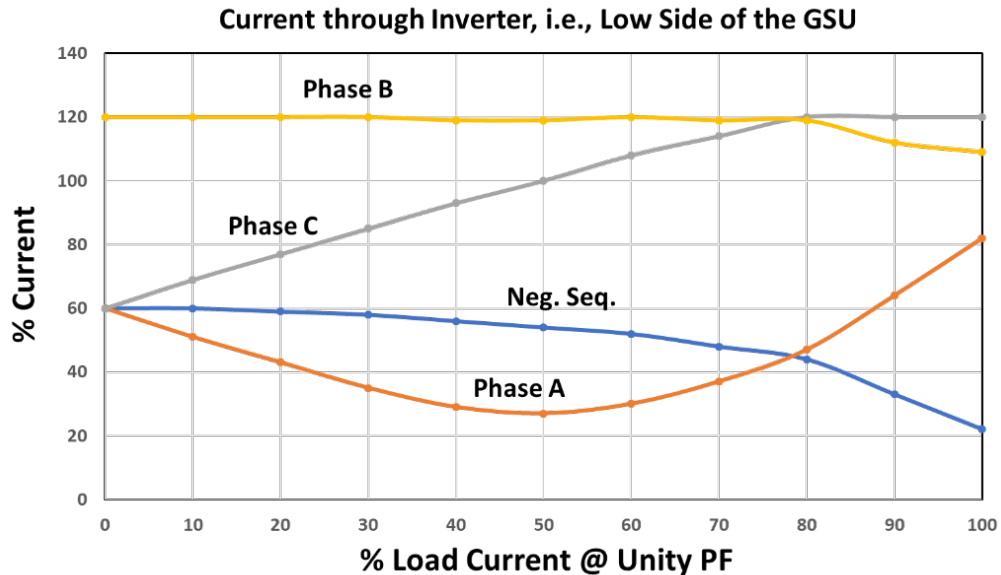


Figure 9: Estimation of current through inverter for close-in (on high-side of unit transformer) line-to-line fault

1.2.4 IBR Zero Sequence Reactive Current Injection

Power electronic sources are typically not grounded so no zero sequence current injection requirements exist. In this regard the IBR is no different than the typical utility-connected conventional generator which supplies only small zero sequence current.

1.2.5 IBR Frequency Ride Through

Nearly all regional transmission regulators and transmission utilities have frequency ride through requirements for interconnecting generators. Figure 10, taken from NERC PRC-024-1 [5], illustrates ride through requirements from different North American regional entities. These requirements apply to all types of generators, conventional sources and IBR, facilities, with the exception in Quebec, Canada where they allow IBRs (asynchronous resources such as photovoltaic and wind turbine generators) to trip instantaneously above 61.7 Hz instead of 66 Hz.

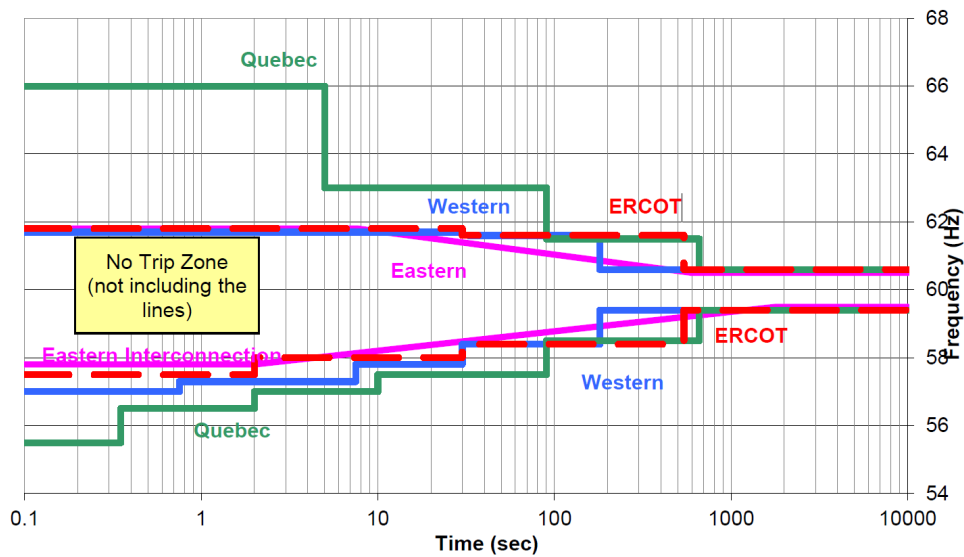


Figure 10: Frequency ride through requirements of North American regional entities

Modern IBR control systems have built-in underfrequency or overfrequency functions which either trip or cease their output during frequency excursions. Depending on inverter settings and implementation algorithm, the IBR may or may not be able to ride through frequency deviations which are within the no trip zone. As an example, nearly 1200 MW of the photovoltaic generation from different facilities in southern California ceased or tripped incorrectly during a high voltage system disturbance due to the inverter control systems at those facilities falsely detecting frequency changes [6].

Modern IBR control systems can be designed to support the power system frequency by adjusting active power outputs in response to system frequency deviations. Similar to conventional generator governor systems, some utility grid codes [7] now require fast inertial response from IBR whose control system must respond by reducing IBR wind

turbine facility output when the system frequency exceeds the nominal or vice versa. A similar strategy could be required for the photovoltaic or battery storage facilities and would have to be developed.

2. IBR – Background and Generic Characteristics

This section provides background on IBRs and their generic characteristics to help understand challenges they present to traditional protection systems.

2.1 Fault Current

To properly evaluate the impact of IBR on the protection system, it is essential to obtain an accurate understanding of the fault current characteristics and contribution levels of various types of IBR. In particular, due to the complex behavior of IBRs during and subsequent to faults, it is improper to simply model them as either a traditional constant voltage source behind Thévenin equivalent impedance (along with a forced current limit) or a basic constant current source at the highest current level.

There are several factors that merit consideration when analyzing the fault current characteristics and/or contribution of an IBR. In the following subsections, some of the major factors are described.

2.1.1 IBR Control Strategy

The fault response of an IBR is fundamentally determined by the control strategy of its power-electronic converter or inverter system, which varies among different manufacturers.

In general, most IBRs are controlled for constant-power output with current limiting functionality, which act as current sources. Therefore, as shown in Figure 11, a very simplified positive-, negative- and zero-sequence equivalent of an IBR has been proposed for short-circuit studies [8][9]. It is, however, important to note that the model of this figure is only suitable for steady-state fault level calculations, and does not meet all protection study requirements.

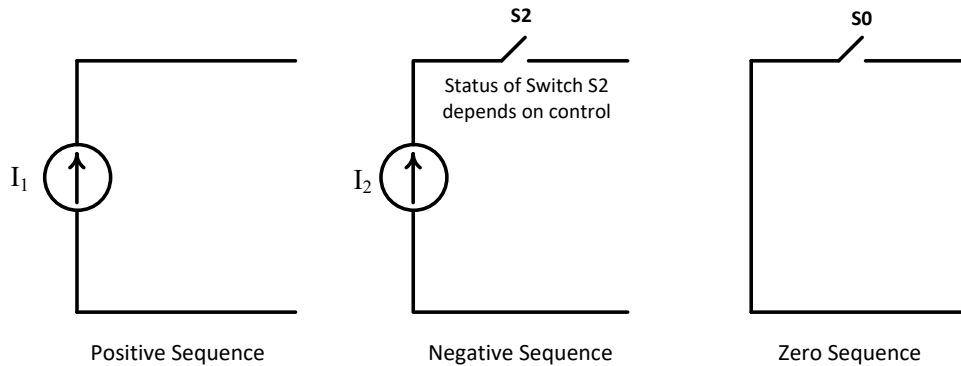


Figure 11: A simplified positive-, negative- and zero-sequence equivalent of an IBR

In most existing designs with full converter interface, the IBR only injects positive sequence current under all operating conditions including balanced and unbalanced faults. This is referred to as coupled sequence control (CSC) scheme [10]. This means that there will not be any negative sequence fault current from the IBR during an unbalanced network fault. Lack of negative sequence current contribution poses a big challenge to traditional protection schemes in transmission systems [11].

With the introduction of new grid codes, as suggested in the previous section and by the German grid code [4], new control designs of the IBR may start contributing negative-sequence current to an unbalanced fault. This control approach is referred to as a decoupled sequence control (DSC) scheme [10] providing an independent control of the positive and negative sequence converter currents. The IBR with DSC control will be able to inject negative sequence reactive current as a function of the negative sequence grid voltage from an unbalanced fault. The injected negative sequence current will lead negative sequence voltage by 90° mimicking a traditional synchronous generator short circuit response with $j1/k$ pu negative-sequence reactance (k being the slope of the characteristic in Figure 8) with current rating limitation. However, as noted before, the magnitude of injected negative sequence current is also affected by the current limiter.

2.1.2 IBR Terminal Voltage

Maintaining constant real power or increasing reactive power during a fault may result in excessive current, particularly if the voltage drop is significant. Thus, the control of an IBR may modify its operating mode to prevent any damage to the power electronic devices. In other words, the fault current contribution of an IBR is a function of its residual voltage (e.g., IBR terminal voltage or collector bus voltage during the fault) [12]. However, the relationship between the inverter current and this residual voltage is nonlinear, which requires extensive testing or high-resolution transient simulation. Detailed IBR control system modeling is required to perform these simulations. Manufacturers and utilities can validate models and then perform simulations to define the voltage-current relationship without divulging control design.

Figure 12 and Figure 13 respectively show typical envelopes of maximum and minimum short-circuit currents as a function of residual voltage for Type IV wind turbine generators (with a full-scale back-to-back power converter as the interface with the grid) and Type III wind turbine generators (with doubly-fed asynchronous generator as the interface to the grid). Typically, voltage greater than 90% is not considered a fault condition. When voltage is less than 20%, most IBRs may cease operation (injection of current). As can be observed, the graphs are represented for different time frames after the fault. Similar curves can be obtained for any other type of IBR.

Another consideration in determination of the IBR residual terminal voltage is the interconnection transformer configuration. Typically, IBRs are interfaced with the host grid through an interconnection transformer that has at least one side in a delta configuration. This configuration will change the fault-induced voltage seen by the IBR measurement schemes installed on the IBR side of the interconnection transformer. Hence, a single-line-to-ground fault on the host grid can appear as a lesser voltage drop on two

phases of the IBR terminal, which would reduce the IBR output current increase as compared to a multi-phase fault.

It is also important to note that the fault current contribution from the IBR is very minimal compared to the contribution of conventional synchronous machine based generating resources, which itself is a major challenge at high penetration levels.

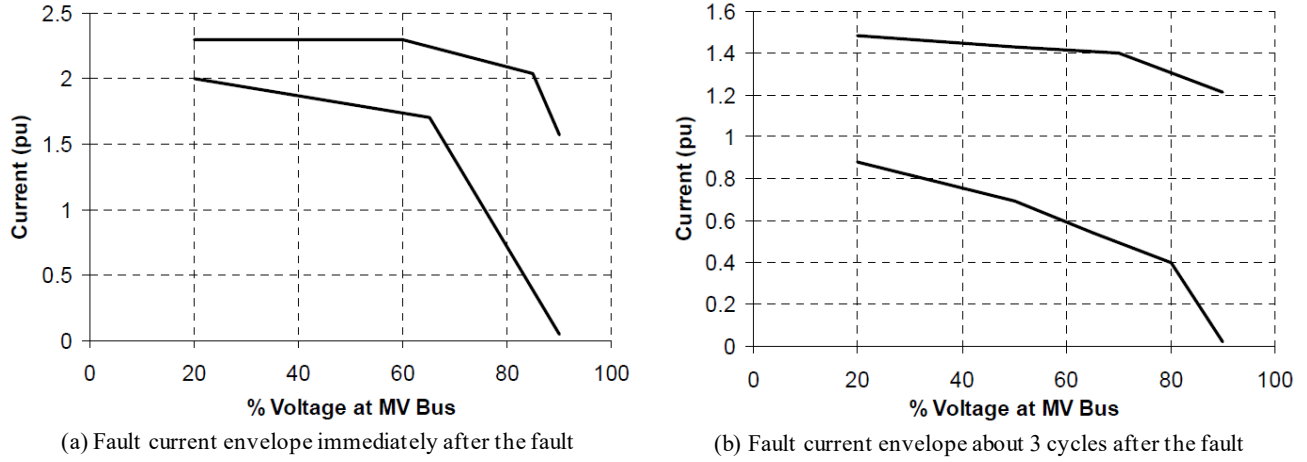


Figure 12: Fault current levels for a Type IV wind turbine generator as a function of residual voltage at the point of common coupling

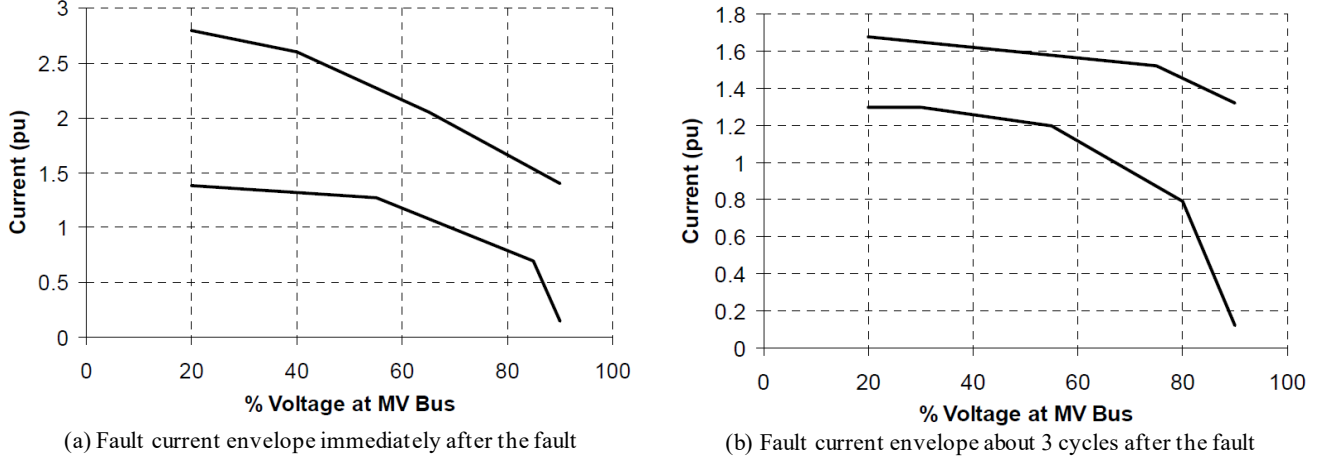


Figure 13: Fault current levels for a Type III wind turbine generator with respect to IBR residual voltage

2.1.3 Pre-Fault Operating Conditions

The IBR fault current contribution, particularly during the first few cycles following the fault, depends on the pre-fault operating conditions of the IBR, e.g., active and reactive power output levels, power factor, control set-points, pre-fault voltage, and input source condition (solar radiation, wind speed, etc.). Therefore, the analysis typically includes the

full range of possible pre-fault operating conditions. However, since the search space of the operating conditions is quite large, a suitable approach is to calculate a maximum possible range of fault current. Generally, the magnitude of fault current contribution could vary from a minimum of zero to a maximum inverter current rating, depending on the pre-fault operating condition.

2.1.4 Time Frame Following the Fault

The fault current of IBRs can generally be studied in two time periods following the fault [13]:

- Before fault detection by the IBR (one to two fundamental frequency cycles after the fault inception): During this period, the residual terminal voltage and the pre-fault operating condition impact the fault current characteristics. Therefore, the IBR output current usually increases (in a range between 1.1 pu to 2.0 pu, after a potential short spike [14]) to maintain the output power constant at a lower-than-rated voltage.
- After the fault detection by the IBR: During this period, the inverter control logic determines the fault current characteristics of the IBR. However, as a general trend, the output current (including relationship between active and reactive current) is similarly increased and maintained at a level corresponding to the voltage drop at the IBR terminals.

Usually, the response of most IBR control systems is fast enough to consider a constant-current output for the entire fault duration. However, the manufacturer may be asked to confirm that the fundamental-frequency fault current can be assumed constant during the fault. Following confirmation, a manufacturer's curve is produced showing the impact of the variations in IBR terminal voltage on fault current contribution. Additionally, the manufacturer also provides the fault current contributions in multiple time frames (e.g. subtransient, transient, synchronous).

For converters with islanding capability (e.g., battery energy storage system), some design specifications require transient capabilities up to 2-3 times rated capacity of the unit for 5 to 10 seconds to support stand-alone applications.

2.1.5 Severity of the Fault

The fault current behavior of some IBRs that do not have a full-scale converter is relatively more complex than the IBRs with fully-rated converter (inverter) system. For example, in a Type III wind turbine generator, the nonlinearity of the crowbar system results in a complex fault behavior. The non-linear crowbar circuit is typically activated on severe faults or faults close to the generator. Upon activation, it shorts the rotor windings through a resistive circuit to protect them from overcurrents or damaging DC link overvoltages. Depending on the fault severity, there are three different classifications of fault current behavior for Type III wind turbine generator:

- Severe faults: The crowbar system constantly operates during the fault and, thus, the fault current response is similar to a simple induction machine. In addition, the fault current can include significant off-nominal frequency components affecting response of relays [15].
- Faults with medium severity: The crowbar system operates for a period of time during which the fault current response is similar to an induction machine. Once the crowbar system is removed, the IBR may transition to a current-limiting mode. The current limit, in turn, may be fixed or time dependent based on the terminal voltage drop.
- Faults of low severity: The crowbar system is not expected to operate. The generator response to fault is like an induction generator for the first few cycles and then the software controlled inverter dominates the current from the generator similar to an IBR with a full-scale converter system.

Modern designs are capable of avoiding crowbar for most faults by momentarily discharging the dc link capacitor between the rotor-side and the grid-side converters.

2.1.6 Sequence Components of the Fault Current

The short-circuit analysis of conventional rotating-machine-based resources assumes that the performance of positive- and negative-sequence networks are fully decoupled, which is a fundamental assumption of symmetrical component analysis. The behavior of IBRs, however, is different from that of conventional rotating-machine-based resources as they usually attempt to maintain balanced current, even under unbalanced faults.

The phase-angle of the fault current can be approximated based on the utility fault ride through requirements. Grid codes define reactive current injection as a function of residual voltage, but do not normally specify values for active current injection during faults. Thus, some assumptions are required to define the current phase-angle. The amplitude and angle of the fault current can also be specified by the manufacturer, if such level of detail is necessary for fault studies.

2.1.7 Power Electronic Transient Rating

Most of the inverter manufacturers, particularly energy storage system vendors, provide both continuous and transient ratings for their products. Since the current contribution of the power electronic converter is in a typical range between 1.1 pu to 2.0 pu, the nominal and transient rating of the converter will determine the maximum fault current magnitude of the IBR. The transient overcurrent of the power converter allows higher fault capacity for adequate duration to support the protection system.

2.1.8 Power Electronic Protection

There are typically conflicting requirements between the power electronic protection schemes and the utility low voltage fault ride through requirements. For distribution interconnections, the utility requires the IBR to disconnect from the electric grid within a

specified period of time after occurrence of a fault [16]. Therefore, the IBR protection can affect the level and time frame of the IBR currents when a fault occurs. In addition, the power electronic protection elements may qualify a system fault and disconnect (or block the currents) because of incorrect frequency measurement and/or sensitivity to harmonics induced by the fault, particularly in weak systems. Thus, evaluating frequency protection elements for the power electronic devices as a part of protection studies is valuable.

3. Traditional Line Protection Relaying

Distance and directional overcurrent protection schemes are widely used in transmission line protection relaying. High penetrations of large capacity IBR facilities into the transmission system are now introducing new challenges to these traditional protection schemes. In this section a background discussion on polarized distance and negative or zero sequence polarized directional overcurrent relays will help to understand how low-magnitude short circuit current with dynamically changing internal impedance (and angle) of an IBR adversely impacts their reliability.

3.1 Distance Relay

Many utilities apply mho or Quad characteristics of distance relays for transmission line protection applications. Self-polarized mho characteristic is used as an example to discuss the influence of IBR on the distance relay.

Self-polarized mho characteristic is represented by a circle in the first quadrant of R-X plane and passes through origin as shown in Figure 14(a). The diameter of circle is the reach or “line-of-the-sight” of the relay which is located at the origin and looking towards the line being protected. As an example, it can be a relay located at CB2 in Figure 4 and protecting the line interconnecting IBR and integrated power system. Influence of large changes in source (IBR) impedance, behind the relay or in the third quadrant of R-X plane, is not obvious when viewing the self-polarized mho characteristic passing through origin.

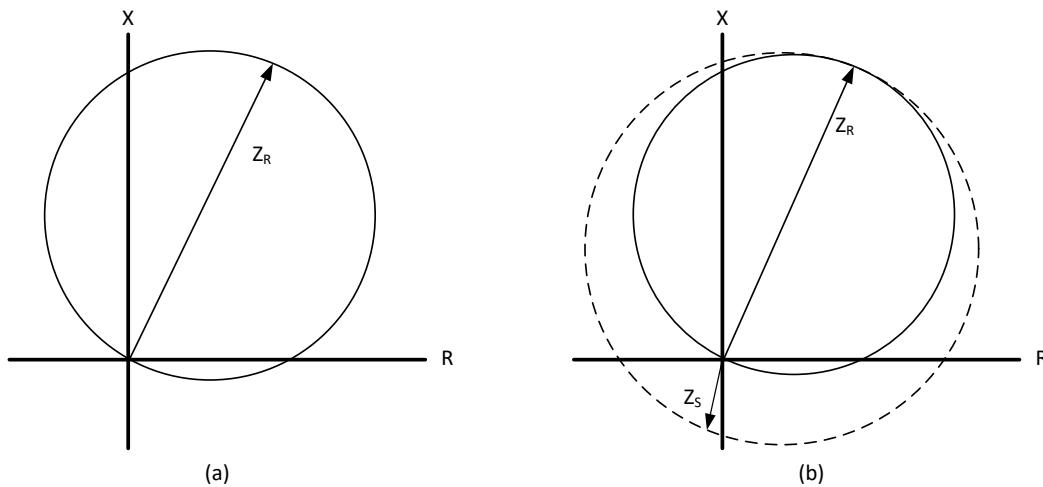


Figure 14: Positive sequence Polarized mho distance characteristics
(a) without and (b) with dynamic expansion

A typical mho relay uses some form of voltage polarization (other than self-polarization) to provide coverage for the faults close to the relay location or near the origin where the self-polarized mho would not produce operating torque. Figure 14(b) illustrates an expanded characteristic of a commonly used positive sequence voltage polarized mho relay [17]. This expansion is proportional to the source impedance behind the relay [18]. Dynamic variations in highly inductive source impedance (magnitude and phase angle) behind the relay (in third quadrant) for a forward fault influence this mho expansion that can cause a distance relay to over- or under-reach. A low short circuit current IBR appears as a high impedance source behind the relay for a forward fault and in-turn significantly expands the mho circle. Since the source impedance depends on the IBR control system, the mho expansion can be anywhere on the R-X plane – not necessarily behind the relay in third quadrant for a forward fault – threatening protection reliability. The non-homogeneous phase angle relationship between an IBR and remote source impedances negatively impacts reliability of a distance relay during unbalanced short circuit involving fault resistances. Most importantly, a distance relay operation is supervised by some minimum phase current. If the IBR output current drops below this minimum value before inherent or intentional added coordination time-delay that the relay takes to operate, the relay will then fail to trip. Depending upon weather conditions, the IBR may not have enough units connected i.e. operating at low capacity prior to developing of fault. Essentially, lack of enough supervising current is a risk to distance relay reliability on the line connecting to IBR.

3.2 Negative or Zero Sequence Polarized Directional Relay

This section provides a brief overview of principles used in conventional negative and zero sequence voltage polarized directional relaying using a conventional source. This background will help later in this report where actual and simulated examples are used to illustrate the reliability risk to negative sequence polarized relays and benefits of zero sequence polarized directional relaying for ground relaying in presence of high penetrations of IBR.

Figure 15 shows an effectively grounded and networked system that has two lines connected in series. These lines tie two systems sourced by traditional synchronous generators. A negative or zero sequence voltage polarized directional relay is shown at Breaker 2 location. This figure has three parts: (a) the top part is a simplified one-line diagram with a solid ground fault in the forward direction of the relay at Breaker 2 location, (b) negative or zero sequence reactance diagram of the network and (c) 60-Hz phasor diagram illustrating the relationships between negative or zero sequence polarizing voltage and corresponding sequence current for the forward fault seen by the relay at Breaker 2 location. Neglecting network resistance for illustrative simplicity, the sequence current leads the corresponding polarizing voltage for the forward fault seen by the relay. Likewise Figure 16 is identical to Figure 15, except the fault direction is reversed for the relay at Breaker 2 location. As a consequence, the relationship between negative or zero sequence polarizing voltage and its corresponding sequence current are also reversed, i.e. the current now lags the voltage for the reverse fault.

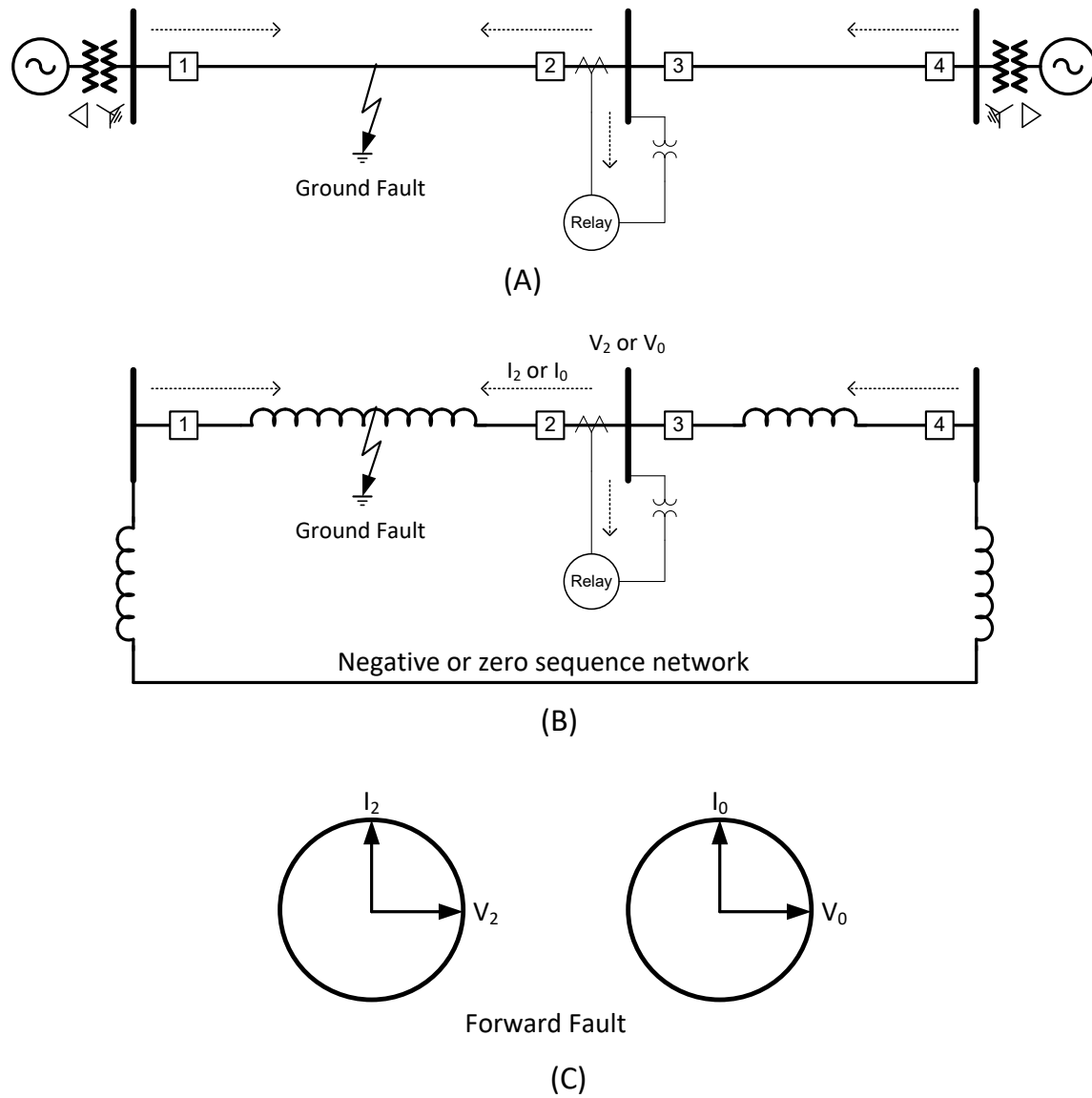


Figure 15: Negative or zero sequence polarized voltage relationship to its corresponding sequence current for a forward fault with conventional sources at both line terminals

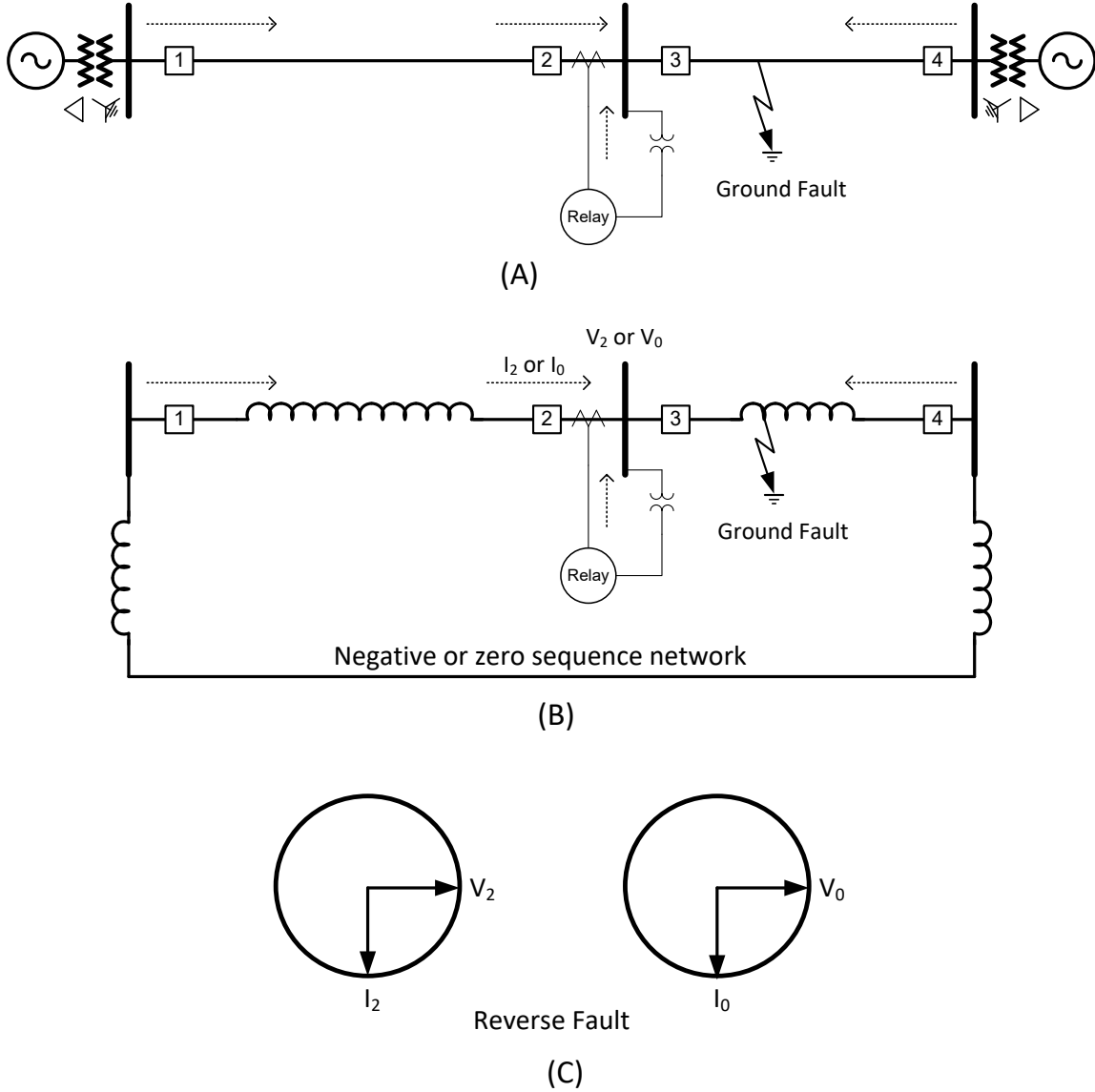


Figure 16: Negative or zero sequence polarized voltage relationship to its corresponding sequence current for a reverse fault with conventional sources at both line terminals

In typical implementations, negative and zero sequence directional elements are enabled only when the respective sequence current is above a certain minimum threshold. This minimum sensitivity increases the reliability of directionality decisions.

4. Challenges to Traditional Line Protection and Solutions

This section highlights challenges and proposes solutions to address them in traditional protection schemes, such as directional ground fault, negative sequence overcurrent, phase distance, differential and power swing protection when used for protecting lines with IBR.

4.1 Negative Sequence Based Directional Ground Fault Relaying

The intent of this section is to discuss the measures in conventional line protection schemes to counter for low-magnitude or uncertainties in characteristic of the short circuit current, in particular lack of negative sequence current, from IBRs during ground faults. It uses three examples of recorded short circuit currents and voltages on the lines supplied by IBRs during actual faults. A fourth example is based on the simulation of an actual network in North America with wind turbine generation.

Example 1: An actual single phase-to-ground fault evolving to a double phase-to-ground fault supplied by large Type III wind turbine facilities

Example 2: An actual single phase-to-ground fault supplied by a large solar generation facility

Example 3: An actual single phase-to-ground fault where fault current contribution from Static Synchronous Compensator or STATCOM source contributed to undesirable line protection operation for an out-of-zone fault

Example 4: A simulated phase-to-phase fault supplied by a Type IV wind turbine generator facility

The report uses only three actual faults but several other faults confirmed the findings.

4.1.1 Example 1 – Type III Wind Turbine Generator

Type III wind turbine generator stators are directly connected to the grid as shown in Figure 3. The generator rotors are built with three-phase windings which receive their excitation from a power converter also connected to the grid. Since the converter system is only interfaced to the rotor circuit and the stator is directly connected to the grid, the power converter contributes only partial i.e. 30-40% of the generator rated power. The laminated rotor with short time flux time constant and power electronic excitation system provides fast reactive power control and voltage regulation.

System Description:

Figure 17 shows the 230 kV single circuit transmission system between the integrated system of utility at station (ISTN) and a remote transmission station (RTLRL).

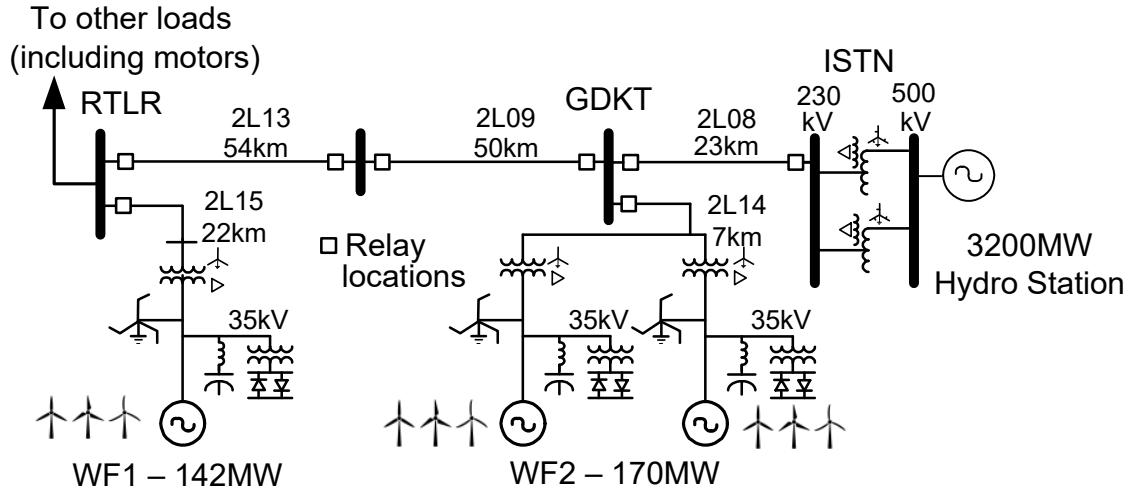


Figure 17: System One-line diagram – Example 1

ISTN is connected to a large hydro generating station. RTL supplies power to transmission and distribution customers. RTL is also POI for the 142 MW Type III wind generating station WF1 (Wind Farm 1) which is an independent power producer or an IPP. GDKT Terminal is a switching station that interconnects another IPP of 170 MW Type III wind generating station WF2 (Wind Farm 2). Both generating stations were integrated into the grid via wye-grounded/delta transformers with wye-grounded windings on the 230 kV. This transformer configuration not only provides effective grounding to the transmission system but also acts as a source of zero sequence current for ground faults on the transmission system. In protection literature, these transformers are also referred to as zero sequence current sources.

Ground Short Circuit Analysis:

On 5th August, 2014, 2L08 experienced a Phase B-to-ground fault about 13.5 km from ISTN. After about three-cycles, it evolved into a Phase B-to-C-ground fault. Figure 18 shows “filtered” records captured by ISTN 2L08 relay looking into the fault. “Filtered” refers to analog signals that were preprocessed by the 60-Hz band-pass digital filter in the relay. In this figure: the top analog traces are phase currents; top middle analog traces are magnitudes of the three sequence currents; bottom middle digital trace is status of the negative sequence forward directional (32GF) element in the multifunction microprocessor-based relay; and the bottom phasor diagrams are phase angles of each sequence current relative to its corresponding sequence voltage at about two cycles into fault (see cursor indicating “phasor time reference” on the figure). The positive sequence voltage is used as the reference and phasors rotate counter-clockwise. Zero and negative sequence current contributions were substantial from the strong integrated system. Their phase angles were leading their respective sequence voltages by about 90° as expected in the conventional synchronous system with inductive path to a forward short circuit fault. This phase angle remained consistent even after the fault transitions from Phase B-to-ground to Phase B-to-C-to-ground and did not change until after the fault was cleared. The negative sequence forward directional element asserted shortly after the fault inception and remained asserted until the fault was cleared.

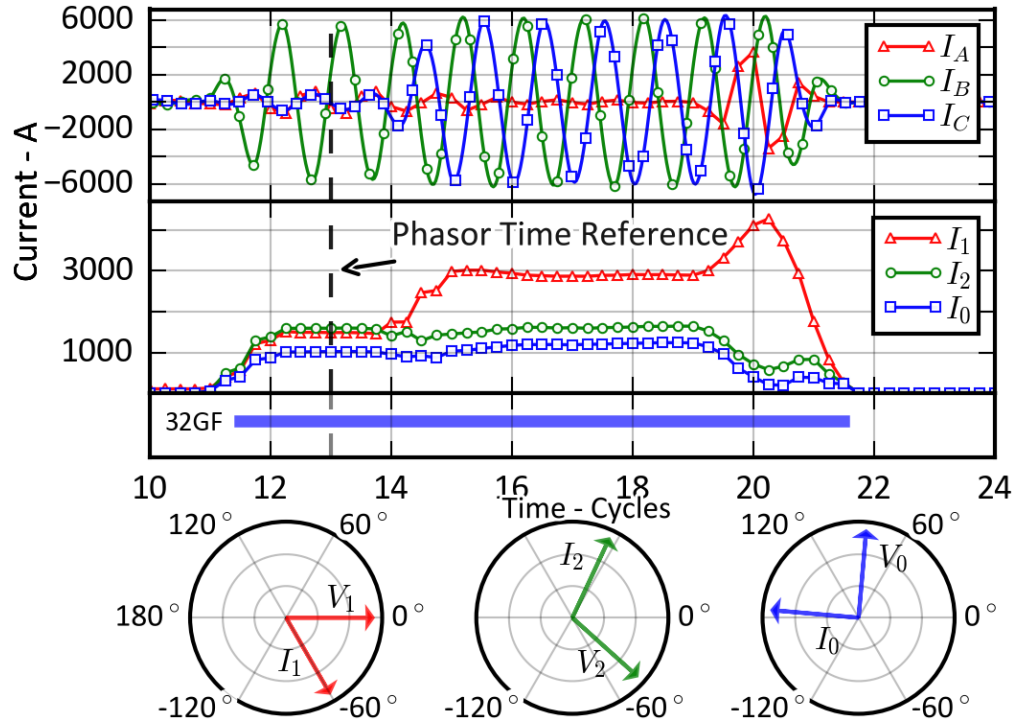


Figure 18: ISTN 2L08 relay records for an evolving forward ground fault contributed by conventional synchronous source.

Figure 19 shows records from the GDKT 2L08 relay for the forward fault current by the wind turbine generation. Zero sequence current magnitude was more than 600 A and led the corresponding sequence voltage by about 90° while the negative sequence current was less than 90 A and was almost *anti-phase* with the corresponding sequence voltage. The negative sequence forward directional element asserted transiently during the event.

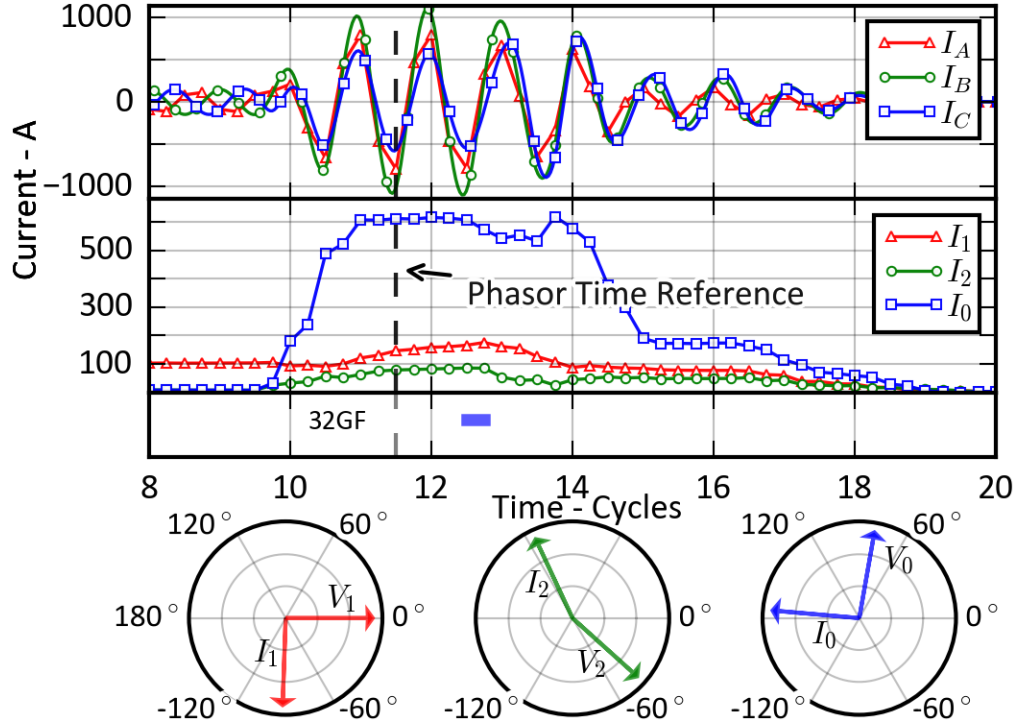


Figure 19: GDKT 2L08 relay records for an evolving forward ground fault contributed by Type III wind turbine generator

Figure 20 shows records from the RTLRL 2L15 relay for the reverse fault i.e. source of the reverse fault current was Type III wind generating station. Initially zero sequence current was 90 A and it increased to about 140 A as the fault evolved from single line- to double-line-to-ground fault. The phase angle of zero sequence current was consistently lagging the zero sequence voltage by 90° . Negative sequence current was less than 40 A and was almost *in-phase* with the corresponding sequence voltage. Though phase angles are shown at about two cycles after the ground fault inception, they maintained similar relation for the duration of the fault. In this figure, the digital trace is the status of the negative sequence reverse directional element (32GR) which picked up only transiently across entire event.

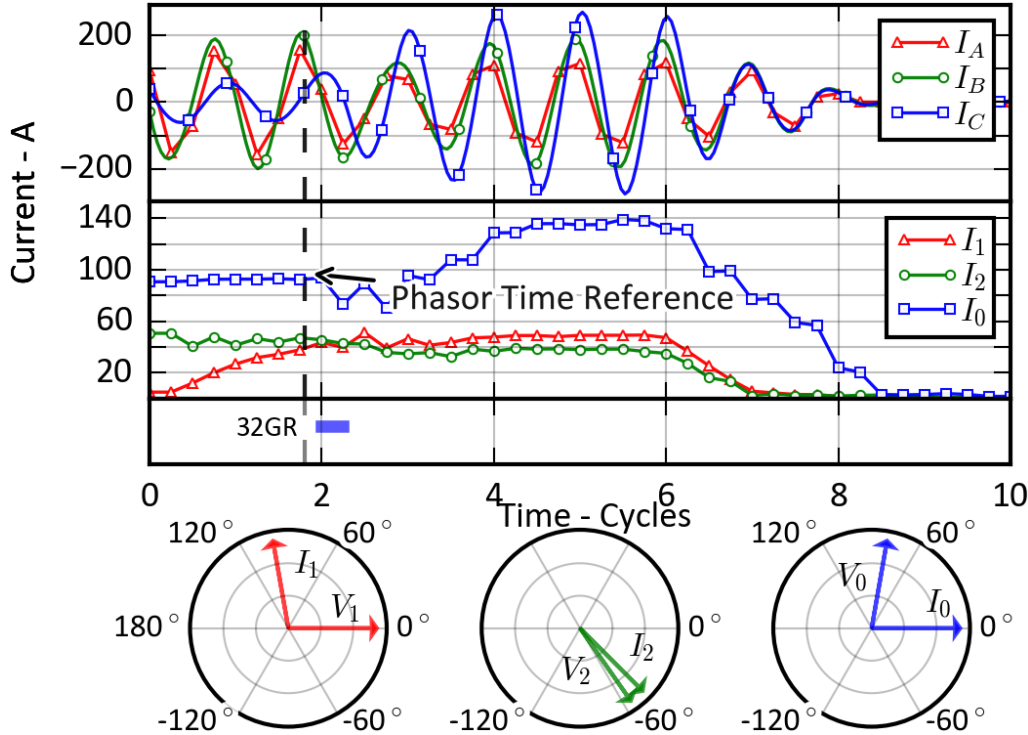


Figure 20: RTLR 2L15 relay records for an evolving reverse ground fault contributed by Type III wind turbine generator

Lessons Learned:

Short circuit current analysis confirmed that the negative sequence current phase angle relationship with the negative sequence voltage from a Type III wind turbine generator is unlike that of conventional synchronous sources and is not readily known. Therefore, a conventional negative sequence current based scheme can't provide reliable directional protection against ground faults in situations with high penetrations of IBR influencing short circuit currents. However, if these generators are connected through a transformer that is a source of zero sequence current, then ground fault protection can be achieved using zero sequence current. The protection will be predominantly independent of the type of wind turbine generator or its associated control algorithm because the interconnecting transformer, not wind the turbine generation, is the source of zero sequence current.

4.1.2 Example 2 – Solar Generation Facility

A solar (photovoltaics) generation facility interconnects to the grid through a DC-AC inverter system. Typically the fault current from the inverter is governed by the inverter control system within two cycles after the short circuit. In the absence of an interconnection or grid code requirement, the control system is often programmed to restrict the magnitude of negative sequence current. If an inverter design permits limited supply of negative sequence current, it may have non-inductive source characteristic angle rendering unreliable in decisions from the devices operating on negative sequence relaying

principles. This section compares line-to-ground short circuits from strong utility sources and solar resources to illustrate that negative sequence relaying cannot be applied on a line connecting a solar facility producing negative sequence in insufficient magnitude and/or phase angle.

System Description:

Figure 21 shows a simplified 230 kV regional one-line diagram of a utility system. In this system, Westley and LB substations are connected to the integrated system having synchronous sources with low short circuit impedance. A 100 MW solar generation facility is connected to Quintito substation through a dedicated 230 kV transmission line. This facility has a three-winding 230/35 kV step up transformer whose configuration is wye-grounded/wye-grounded with delta tertiary serving as source of zero sequence current bypassing the inverter control system.

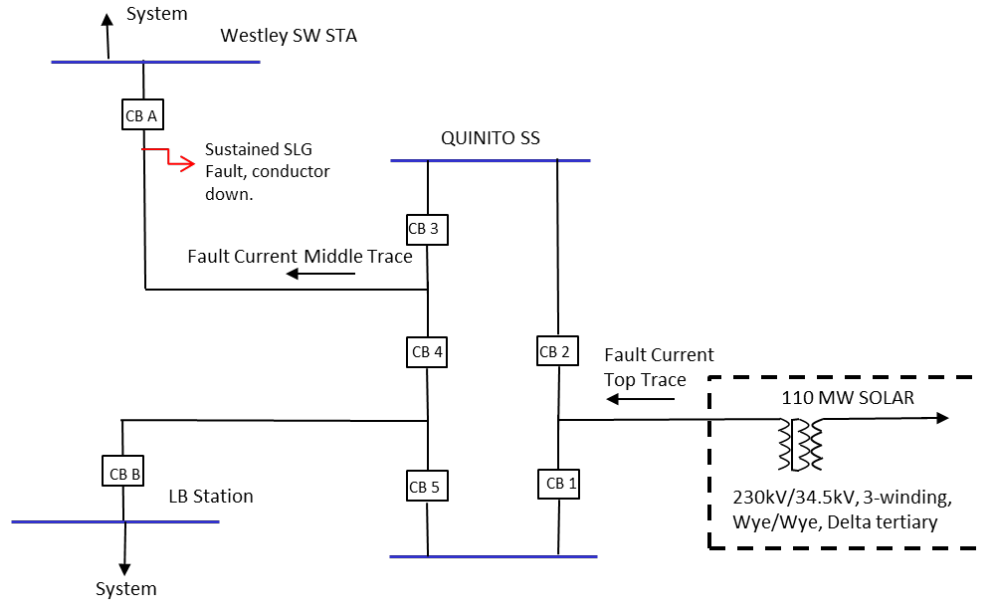


Figure 21: System One-line diagram – Example 2

Ground Short Circuit Analysis:

A permanent Phase B-to-ground fault was experienced on the transmission line between Westley and Quintito substations. Figure 22 illustrates dissimilar natures of the short circuit currents supplied by the integrated system and the solar generation facility. The upper traces are captured waveforms during the event by the relay located on the 230 kV side of interconnecting transformer of solar facility. Likewise, the lower traces are captured waveforms by the line protection relay located at CB 3 and CB 4 at Quintito substation. The line protection correctly saw an internal transmission line fault whose waveform characteristics were identical to a traditional single phase-to-ground fault because it was receiving short circuit current contributions dominantly from the conventional sources within the integrated system. The line protection relay isolated the fault by tripping CB-A, CB-3 and CB-4. The upper traces illustrate that the solar facility rode through the low-voltage caused by a short circuit event external to the line connecting the facility to Quintito.

However, the short current characteristic did not resemble traditional single phase-to-ground fault current because of restricted supply of negative sequence current by the solar generation facility. The negative sequence contribution was only approximately 21% of the positive sequence current and 24% of the zero-sequence current, indicating the inverters were limiting their negative sequence current injection in response to the fault.

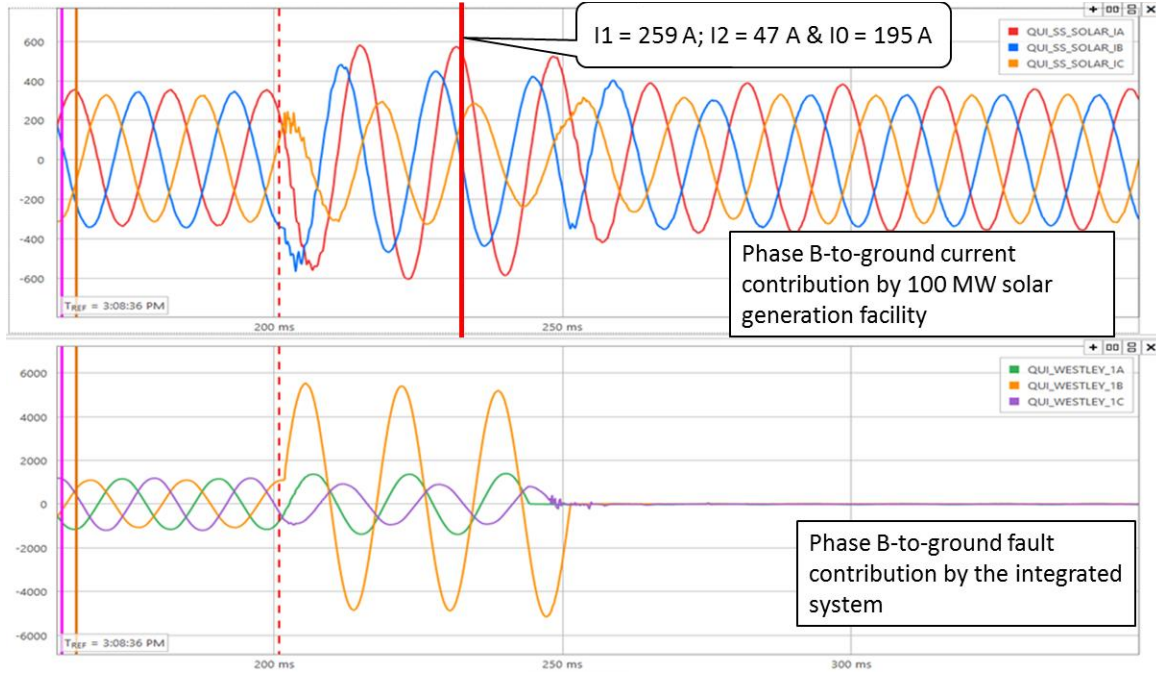


Figure 22: Relay records for a single phase-to-ground fault contributed by the solar generation facility and integrated system

Lessons Learned:

In the absence of an interconnection grid code, the inverter control system of solar generation facility will likely restrict the magnitude of negative sequence current during unbalanced faults. As a consequence, the negative sequence relaying cannot be reliably applied on a line connecting a solar facility with this characteristic.

4.1.3 Example 3 – STATCOM

STATCOM sources interface with the grid via a full-scale power converter. Its output is fully defined by the power converter control instead of natural response of the source to sudden changes in power system. The following is a discussion of an undesirable line protection operation during a ground fault caused by a STATCOM whose control is designed to respond to supply positive sequence reactive current for all symmetrical and unsymmetrical faults somewhat similar to characteristic shown in Figure 7. Thus, its positive sequence impedance is viewed to dynamically change with voltage depression. Since it tries to suppress negative sequence currents during the unbalance, its negative sequence impedance can be viewed as very high but dynamically changing with changes in voltage depression.

This section compares the currents seen by two 138 kV transmission lines, referred to as upstream and downstream lines, feeding a single phase-to-ground fault. The downstream line had a stiff synchronous source from the integrated system whereas the upstream line had a relatively weak source and large STATCOM. The upstream line experienced an undesirable protection operation and the fault records are used to identify the cause of this operation.

System Description:

Figure 23 shows a simplified one-line diagram of the 138 kV Regional Transmission System in a utility system. ISKL is a transmission switching station connected to the integrated grid. Upstream of ISKL, 1L10 is a 195 km long transmission line to TAVL which connects to a transmission system interconnecting several small synchronous sources of generation. There are several transmission tap-connected pumping loads along the transmission line, not shown on the one-line diagram, from ISKL to TAVL. A STATCOM, continuous rating of ± 12 MVar with temporary overload rating of ± 24 MVar, is installed at TAVL for dynamic reactive support during voltage depressions from transmission faults. Per unit Thévenin impedances of the synchronous sources (ignoring STATCOM contribution) behind ISKL 1L41 and TAVL 1L10 terminals are listed in tables below the figure. The positive sequence source strength behind ISKL is about ten times stronger than behind TAVL. The zero sequence source strength at TAVL is comparable to ISKL largely because the STATCOM transformer is 138 kV wye-grounded and delta on low-voltage reducing the zero sequence impedance at that station.

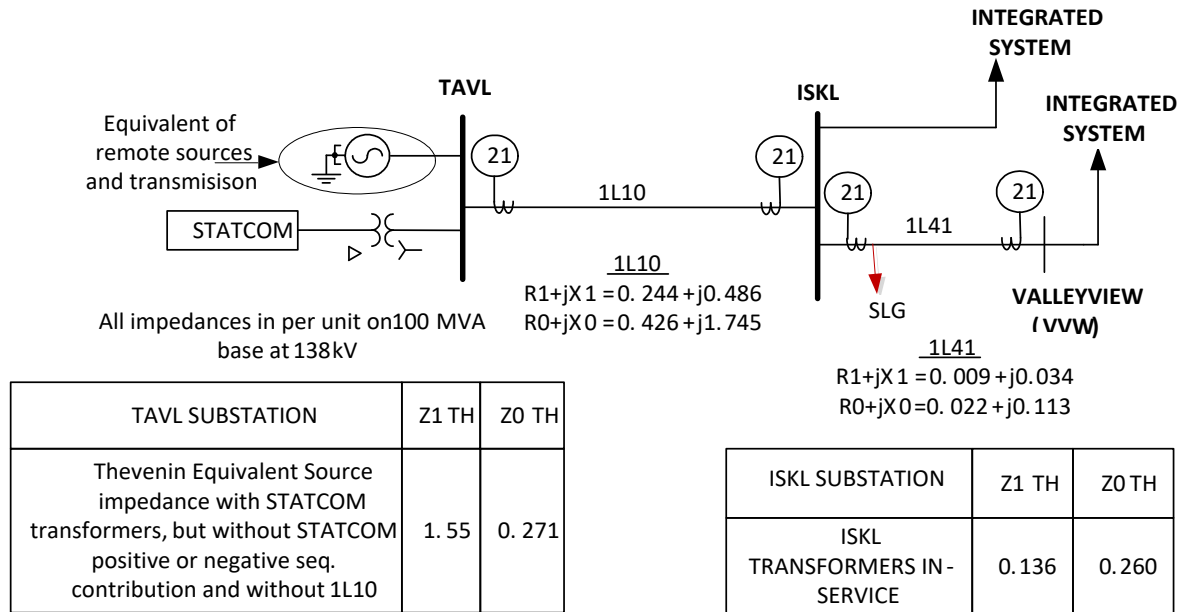


Figure 23: System One-line diagram – Example 3

Ground Short Circuit Analysis:

On 30th June, 2015, 1L41 experienced a Phase C-to-ground fault close to the ISKL substation. During the fault, instantaneous Zone 1 ground quadrilateral distance protection (Z1G) operated correctly at ISKL 1L41 but incorrectly at TAVL 1L10. Because of TAVL 1L10 over tripping on an external line fault, the entire regional system lost the supply. To illustrate the cause of over tripping, the fault currents seen by the two line protection systems are analyzed and compared with those obtained from the system short circuit model that did not include STATCOM at TAVL.

Figure 24 shows “filtered” records captured by ISKL 1L41 relay looking into the fault, except the digital traces are statuses of Zone 1 ground quadrilateral distance (Z1G) and trip (TRIP) elements. In Figure 24, the phase angle of each sequence current relative to its corresponding sequence voltage is shown at about 1.5 cycle into the fault. As a result of the ISKL substation being connected to the strong integrated system, the ground fault current was about 2000 A and matched closely with the fault current obtained from the short circuit model. Again as anticipated for the conventional system, the phase angles of the negative and zero sequence currents were leading their respective sequence voltages by 90° throughout the fault event. The zone 1 ground distance element picked up shortly after fault initiation and tripped correctly.

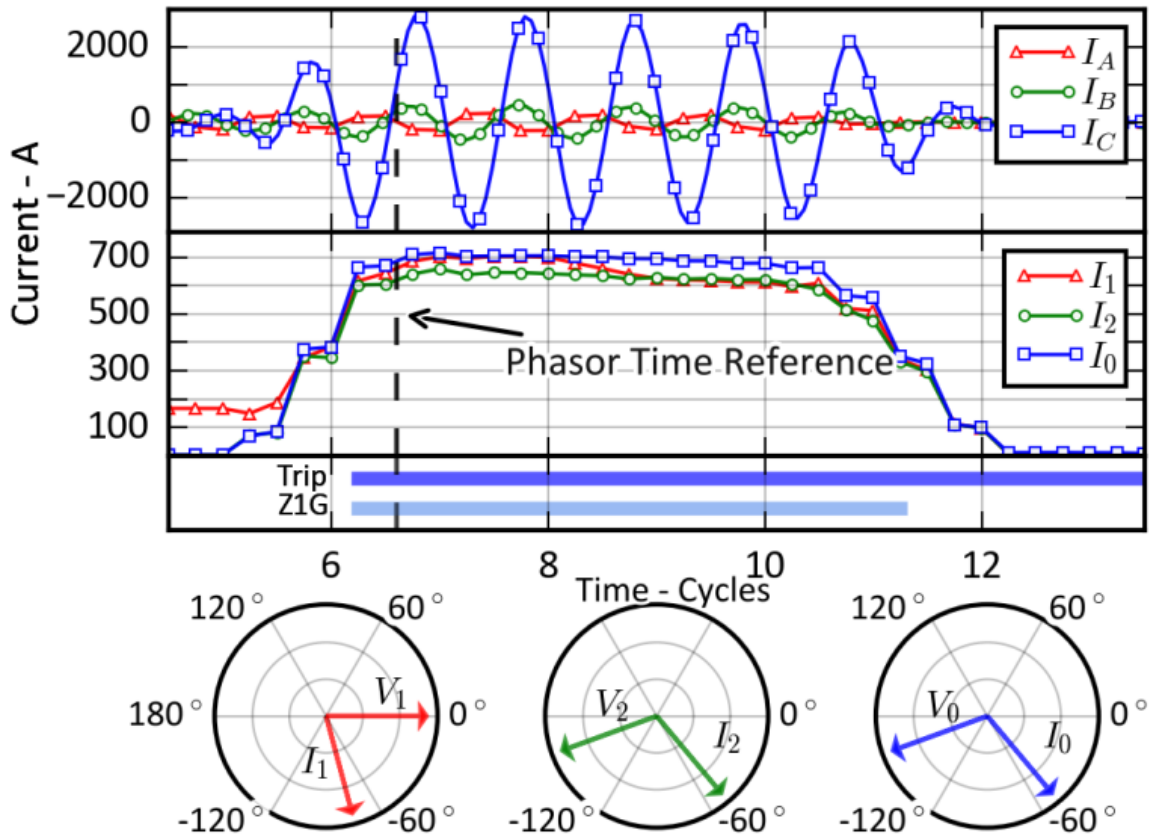


Figure 24: ISKL 1L41 relay records for a forward ground fault contributed by conventional generator source

Figure 25 shows relay records for the same fault from the TAVL 1L10 ground distance relay which operated undesirably. Using Thévenin impedances of the synchronous sources (not taking into account the STATCOM) at TAVL, the maximum rms value of the fault current was expected to be about 200 A but the actual maximum fault current recorded was about 240 A. The positive sequence current was not steady and reached about 130 A which was considerably higher than the 50 A expected from the short circuit model. This higher current was due to reactive current injection from the STATCOM trying to maintain the voltage at TAVL indicating the dynamically changing nature of the STATCOM positive sequence impedance. The negative sequence current had a varying magnitude with the maximum value close to 55 A provided by the model – indicating the dynamic nature of relative high negative sequence impedance of STATCOM. The zero sequence current remained steady at 100 A throughout the event and matched closely with the model because of wye-grounded delta windings of STATCOM and the other upstream transformers. Unlike ISKL, the phase angle of the negative sequence current relative to the respective sequence voltage was varying across the event. At about 1.5 cycles into the fault, negative sequence current was *anti-phase* (180° out of phase) with respective sequence voltage. However, the zero sequence current consistently led the corresponding voltage around 90° during the entire fault event i.e. similar to ISKL.

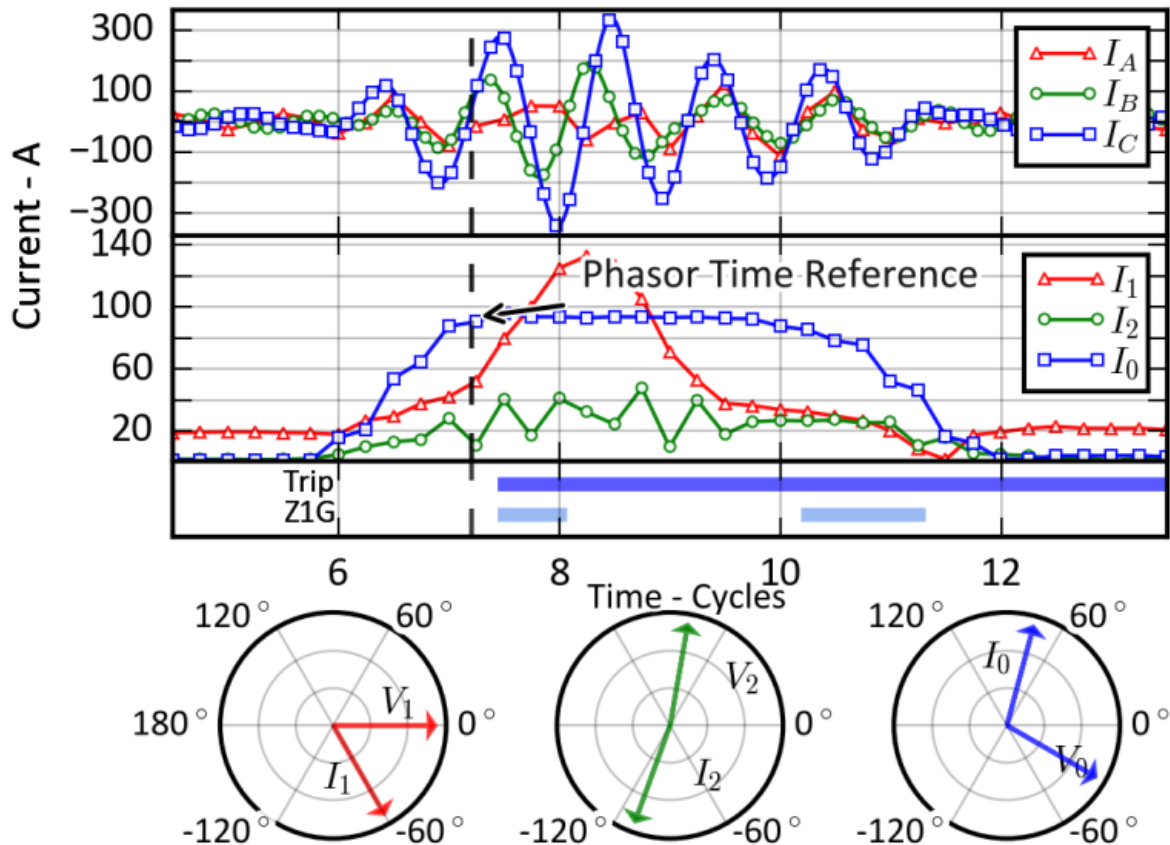


Figure 25: TAVL 1L10 relay records for a forward ground fault contributed by weak conventional source system and STATCOM

1L10 Protection Misoperation Analysis:

The reactance blinder of the ground quadrilateral distance element applied was polarized with negative sequence current. The line impedance is 5.18 secondary ohms (104 primary ohms) and the 21G reactive reach was set to 3.8 secondary ohms i.e. the element overreached and operated well outside of its setting reach. Using the four samples per cycles phasor data extracted from the event captured by TAVL 1L10 relay and its reactive reach characteristic [19] as described by the equation below, the ground quadrilateral distance element response was simulated for Phase C-to-ground fault.

$$X_{CG} = \frac{\text{Imag}(V_C \times I_p^*)}{\text{Imag}(Z_{1ang} \times (I_C + k_0 \times I_0) \times I_p^*)} \quad (1)$$

In Equation 1 above, X_{CG} is the calculated reactance, V_c is Phase C-to-ground voltage, I_c is Phase C line current, I_0 is zero sequence line current, I_p is either zero or negative sequence polarizing current, Z_{1ang} is positive sequence line angle and k_0 is zero sequence current compensating factor. Using negative and zero sequence currents as the polarizing quantities, Figure 26 illustrates the simulated responses of the ground reactance element for the fault duration. Shortly after, i.e. one cycle of the fault inception, the reach calculations from the negative sequence polarized relay transiently dropped below the threshold and caused the protection misoperation. During the fault, the response of this element was somewhat unsteady since the negative sequence current was less predictable. The reach from the zero sequence current polarized relay was steady and remained above Zone 1 threshold before the circuit breaker started to open.

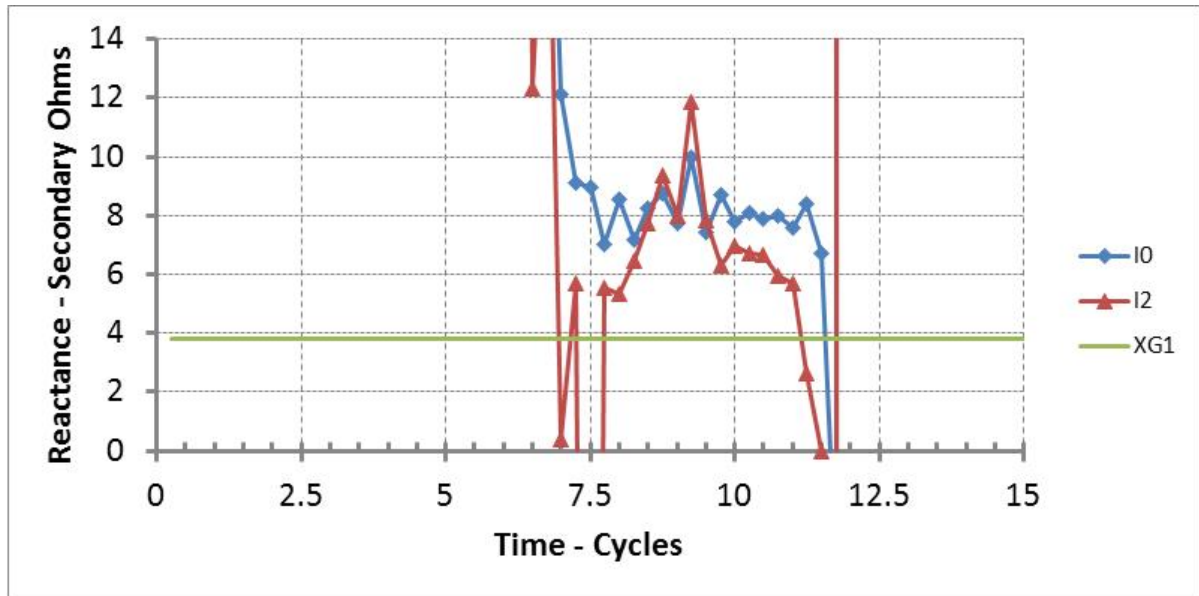


Figure 26: Simulated responses of zero (I0) and negative (I2) sequence current polarized ground quadrilateral distance relays

The misoperation of TAVL 1L10 ground quadrilateral distance relay was attributed to the dynamic response of STATCOM control system immediately after the fault when it was attempting to raise and balance the depressed 138 kV voltage on the faulted phase at TAVL by injecting reactive current behind relay during the ground fault. The negative sequence impedance behind the relay at the TAVL bus was high from the weak synchronous sources. So while STATCOM was attempting to raise and balance the faulted phase voltage, the negative sequence current from the weak source became unsteady and unreliable for use in decisions by the TAVL 1L10 line protection scheme. However, the zero sequence source at the TAVL bus behind the relay was strong due to presence of upstream wye-grounded delta transformers. Therefore zero sequence reactance relay remained steady and unaffected by the STATCOM control during the fault.

Lessons Learned:

Based on the discussion presented in this section and the results shown in Figure 26, it can be concluded that the zero sequence current, instead of negative sequence current, provides reliable ground protection in presence of sources with full-scale power converters contributing to the short circuit currents. The reported event also demonstrated that these sources supply positive sequence current thereby increasing the overall fault current magnitude. But phase overcurrent protection settings are difficult to determine using steady-state short circuit analysis because of unknown contributions from the IBR. Therefore it is important to carefully apply phase overcurrent protection in the presence of these IBR.

4.1.4 Example 4 – Type IV Wind Turbine Generator

This section describes a simulation based study performed on an area of an actual system in North America with a large IBR facility. The purpose is to demonstrate an example of negative sequence directional relay misoperation on short circuit current response of a 300 MW Type IV wind turbine generator facility during a phase-to-phase fault.

System Description:

Full details of an actual utility system, interconnecting five turbine facilities, are provided in Reference [20]. An EMTP RV model of the entire integrated system was built including the detailed representation of each wind turbine generation facility. Figure 27 is a simplified one-line diagram showing only a single wind turbine facility and its interconnection that is relevant to the short circuit example discussed. This facility is comprised of 200×1.5 MW Type IV wind turbine generators whose control system completely suppresses negative sequence current. System details and other wind turbine facilities are hidden for simplicity.

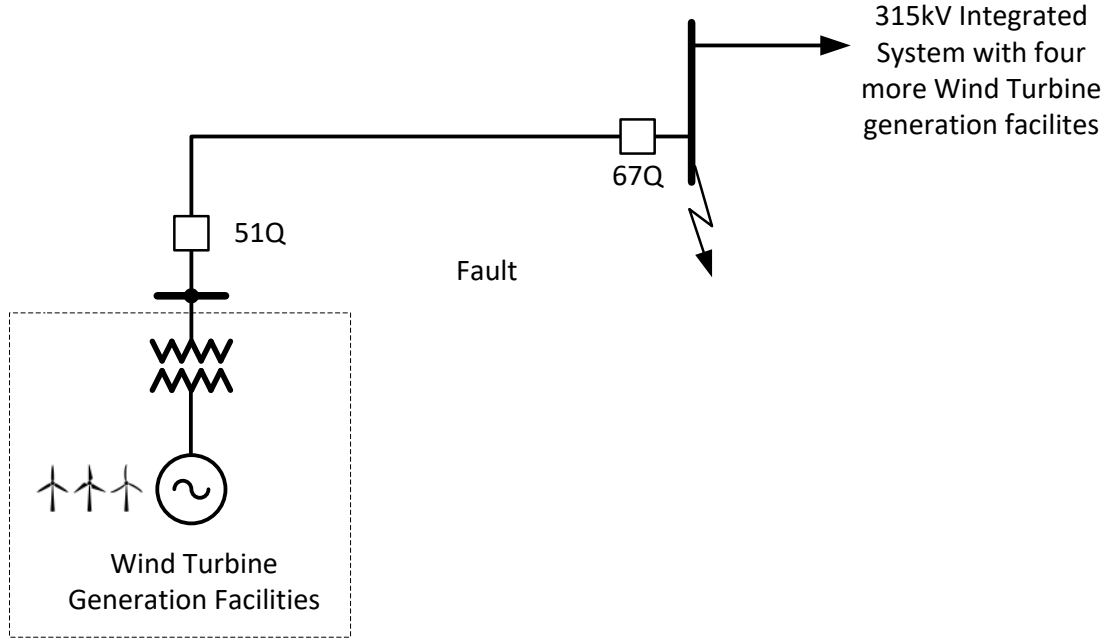


Figure 27: System One-line diagram – Example 4

Simulated Phase-to-Phase Analysis:

A permanent phase-to-phase fault was simulated. It was a reverse fault to the negative-sequence voltage-polarized directional relay (67Q) which was set to look into the line towards the generation facility. The following three scenarios were simulated to illustrate reliability risk of applying negative sequence polarized directional relay on the line interconnecting the IBR facility.

- Scenario 1:** The wind facility as shown in the figure was replaced with a synchronous generator of equal (300 MVA) capacity
- Scenario 2:** The original facility comprising of 200×1.5 MW Type IV wind turbine generators coupled sequence control (CSC) scheme [10], i.e. with no negative current injection control capability
- Scenario 3:** The wind facility comprising of 200×1.5 MW Type IV wind turbine generators decoupled sequence control (DSC) scheme [10], i.e. with negative sequence current injection control capability similar to Figure 8

Figure 28 through Figure 30 show the oscillography data and the response of 67Q under the three scenarios. In these figures, V_2 and I_2 are negative-sequence voltage and current recorded by the 67Q. The 67QF and 67QR represent status of the forward and reverse signals, respectively within 67Q. Negative sequence current threshold 0.02 pu was used to enable the relay; where 1 pu represents facility rated output current 550A.

In Scenario 1 simulating a synchronous source, Figure 28 shows V_2 led I_2 by approximately 88° because of inherently highly inductive nature of the fault current as expected. The relay correctly asserted 67QR and declared a reverse fault. But in Scenario 2, Figure 29 shows

that I_2 was small due to highly controlled response of the Type IV wind turbine generators suppressing negative sequence current. Consequently the relay 67Q response was unstable in that it mistakenly declared a forward fault by asserting 67QF erratically. In Scenario 3, the wind turbine control system, with DSC scheme and slope setting $K=2$, permitted high enough I_2 injection and appropriate angular relationship between V_2 and I_2 contributing to allow correct directionality decision by the relay 67Q,

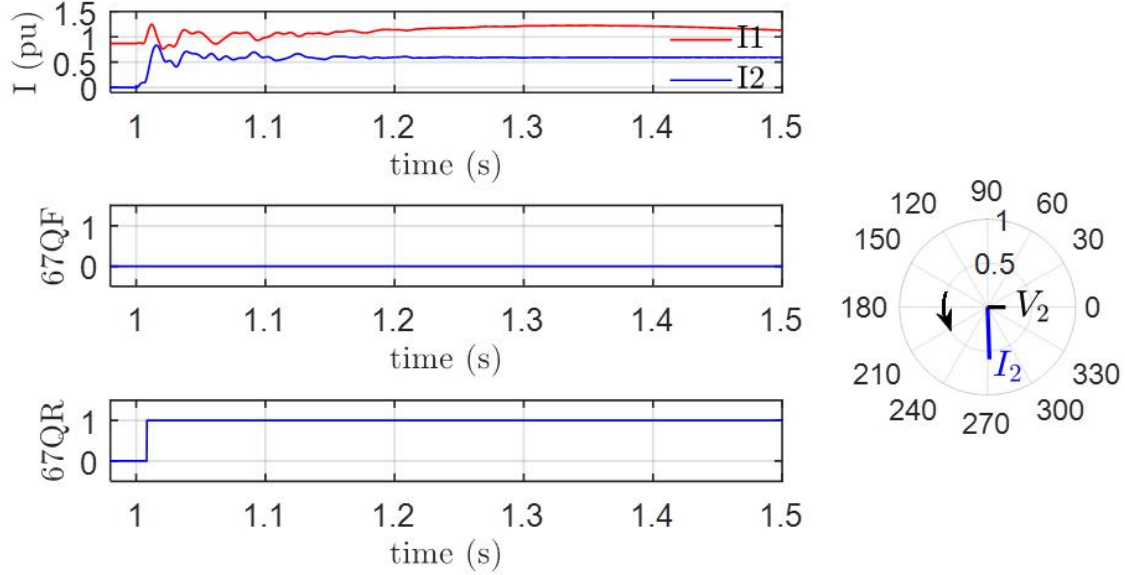


Figure 28: Negative sequence voltage polarized relay responses to simulated phase-to-phase fault with synchronous generator – Scenario 1

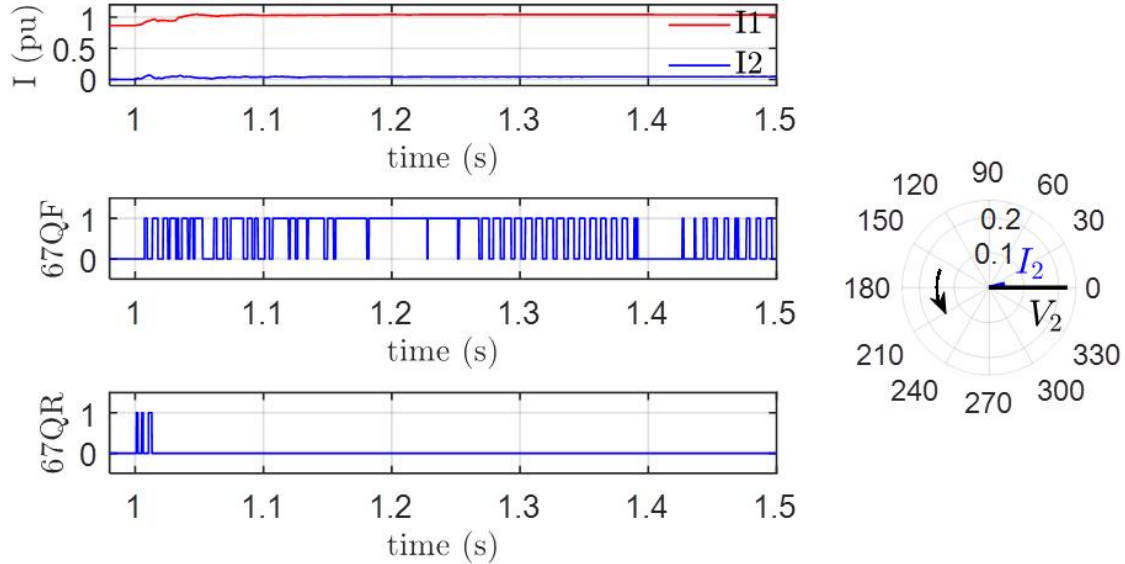


Figure 29: Negative sequence voltage polarized relay responses to simulated phase-to-phase fault with coupled controlled (no negative sequence current injection capability) Type IV wind turbine generators – Scenario 2

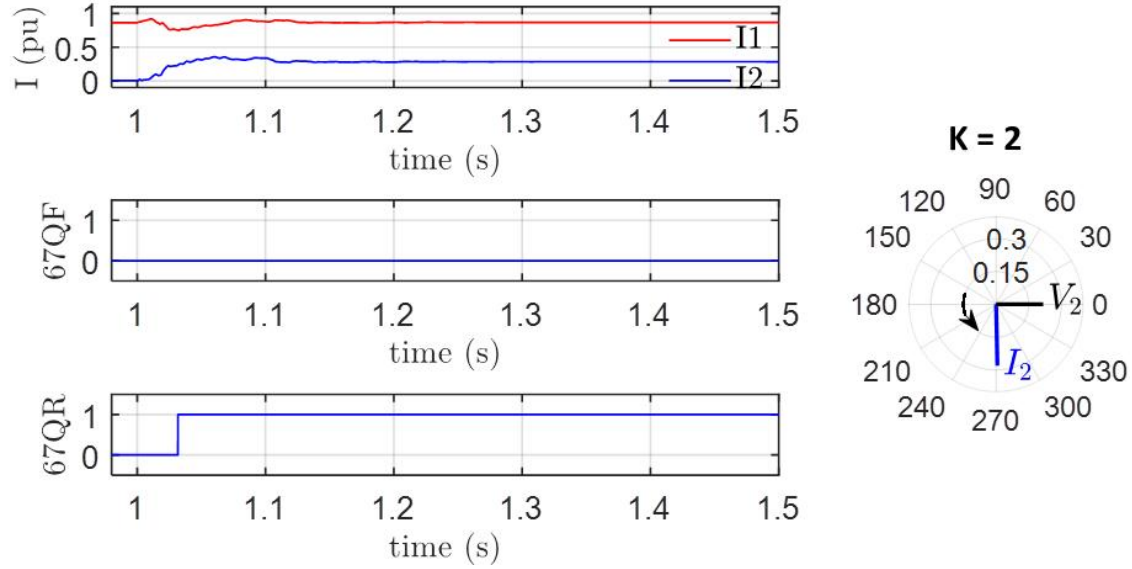


Figure 30: Negative sequence voltage polarized relay responses to simulated phase-to-phase fault with decoupled controlled (negative sequence current injection capability and $K=2$) Type IV wind turbine generators – Scenario 3

Lessons Learned:

Type IV wind turbine generation exhibits a substantially different negative-sequence fault current characteristic compared to synchronous generators. IBR without negative sequence injection capability may negatively impact the performance of negative-sequence directional relay element due to the changed angular relation of negative sequence quantities.

All relaying devices require minimum (settable or non-settable threshold) negative sequence current to maximize reliability of their directional decisions. In the absence of negative sequence current i.e. when it is below minimum threshold, relay behavior can vary from either blocking directional element or converting the directional to non-directional element.

4.2 Negative Sequence Overcurrent Relaying

This section presents the following two examples, field and simulated events, demonstrating that traditional negative sequence overcurrent unreliability when operating on short current supplied by the IBR facilities:

Example 1: A field event involving phase-to-phase-to-ground fault sourced by battery storage and inverter system

Example 2: A simulated event involving phase-to-phase fault supplied by Type III or IV wind turbine generator

4.2.1 Example 1 – Field Event

Photovoltaic or battery storage systems are DC sources that connect to grid through an inverter interface. In absence of grid code, most inverters are programmed to balance their terminal voltage and to suppress their negative sequence current output. Thus, their apparent negative sequence source impedance is typically significantly higher than their positive sequence source impedance. On fault inception, they respond to voltage drop and increase their positive sequence current to a pre-specified current threshold i.e. at their thermal rating. At this level, the inverter almost acts as a current source until the voltage recovers.

Using data recorded for a double line-to-ground fault, this section illustrates that the negative sequence protection cannot be reliably applied when the relay operating current is solely supplied by the IBR. Though the example used is from the distribution system, the lessons learned from this application apply to transmission connected IBR as well.

System Description:

Figure 31 shows the 25 kV distribution feeder one-line diagram and its connection to the storage comprising of 1 MW battery banks which operate between 485 V and 780 V DC. The battery bank connects to a 480 V AC system through a 1.25 MVA inverter which then interconnects to the distribution feeder via a 480V/25 kV step-up delta-wye transformer. The wye winding is grounded on the 25 kV side to maintain effective grounding when the Interrupter is open and the community load is supplied by the stand alone storage system.

The inverter has built-in control functions operating on its output current. The control limits the inverter output to about 3008 A or 2.0 pu of the inverter continuous rating. Upstream of the inverter, an inverse-time overcurrent digital relay (Device 51 in Figure 31) provides coordinated protection with the entrance fuses of customers connected on the distribution feeder.

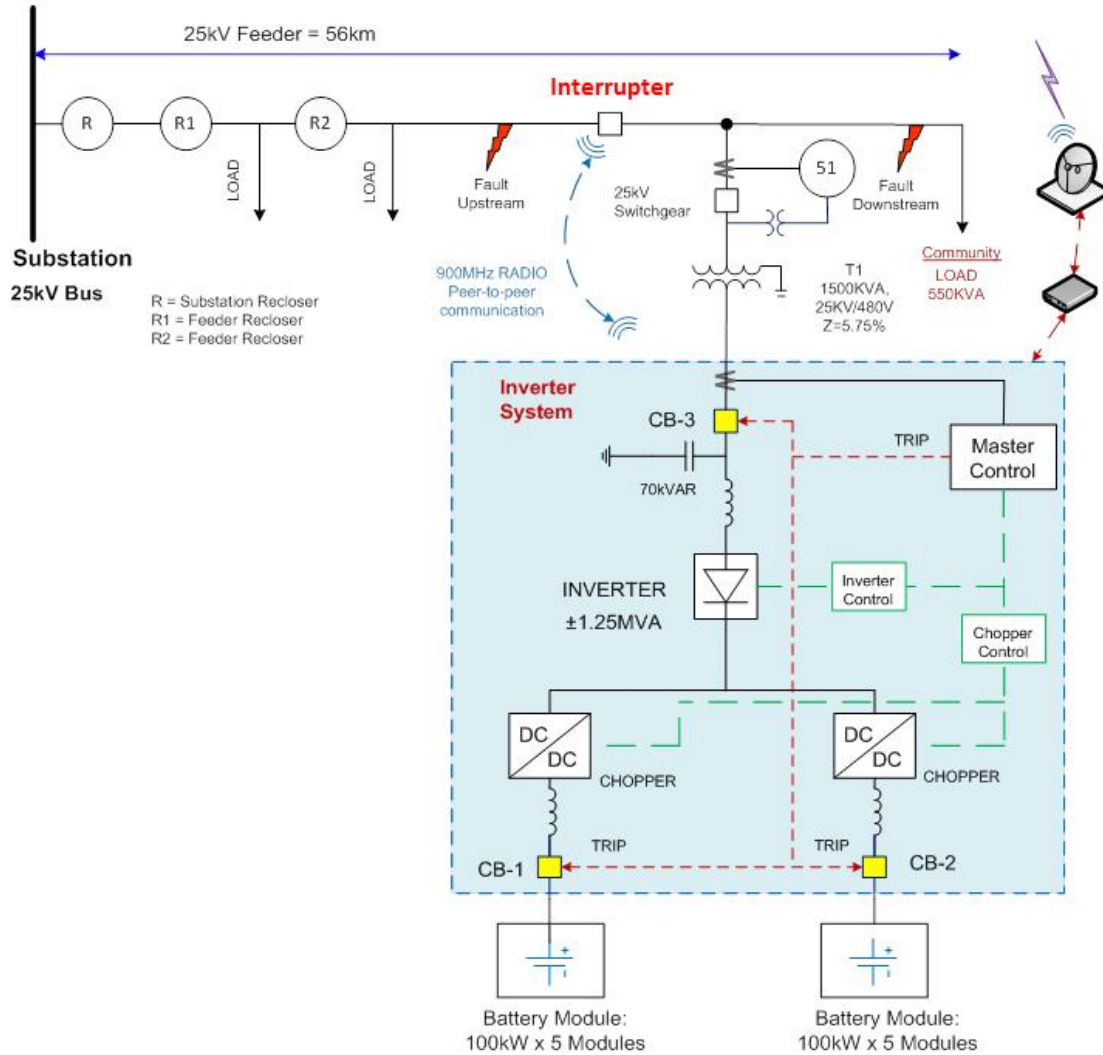


Figure 31: Electrical one-line diagram showing interconnection of DC battery and inverter system

Negative Sequence Fault Current Characteristic

On 27 July, 2017, the distribution feeder experienced a permanent Phase A-to-C-to-ground fault downstream of the Interrupter i.e. on the overhead feeder section between location of battery system and the community load. Feeder protection and inverter protection systems correctly de-energized the feeder. After the Interrupter was opened, the system operator then attempted to restore the community load through the battery system. Due to the fact it was a permanent fault, the feeder re-tripped. Figure 32 shows waveforms captured by the inverse-time overcurrent relay (Device 51) when the feeder was picked up onto the close-in 25 kV A-to-C-ground fault. In the figure, the top trace has 25 kV three-phase-to-ground voltages. Since it was a solid double line-to-ground fault close-in to the relay location, the voltages on the two faulted phases had completely collapsed and the healthy phase had a severe voltage depression. The traces in the upper middle part of the figure are three-phase currents on the 25 kV feeder section, supplied by the inverter through the delta-wye

transformer. Both faulted phases had about 100 A rms current which was limited by inverter control to 3008 A at 480 V ($\sqrt{3} \times 3008 \times 480 / 25000 \text{ A} = 100 \text{ A}$). Traces in the lower middle are magnitudes of three sequence currents. Absence of negative sequence current shows that the inverter source in this application had significantly high negative sequence impedance. The inverter source was connected through a delta-wye-grounded transformer which acted as a strong source supplying high magnitude of zero sequence current. The bottom part of Figure 32 has phasor diagrams. Zero sequence current leads the corresponding voltage by about 95° correctly indicating a forward fault direction.

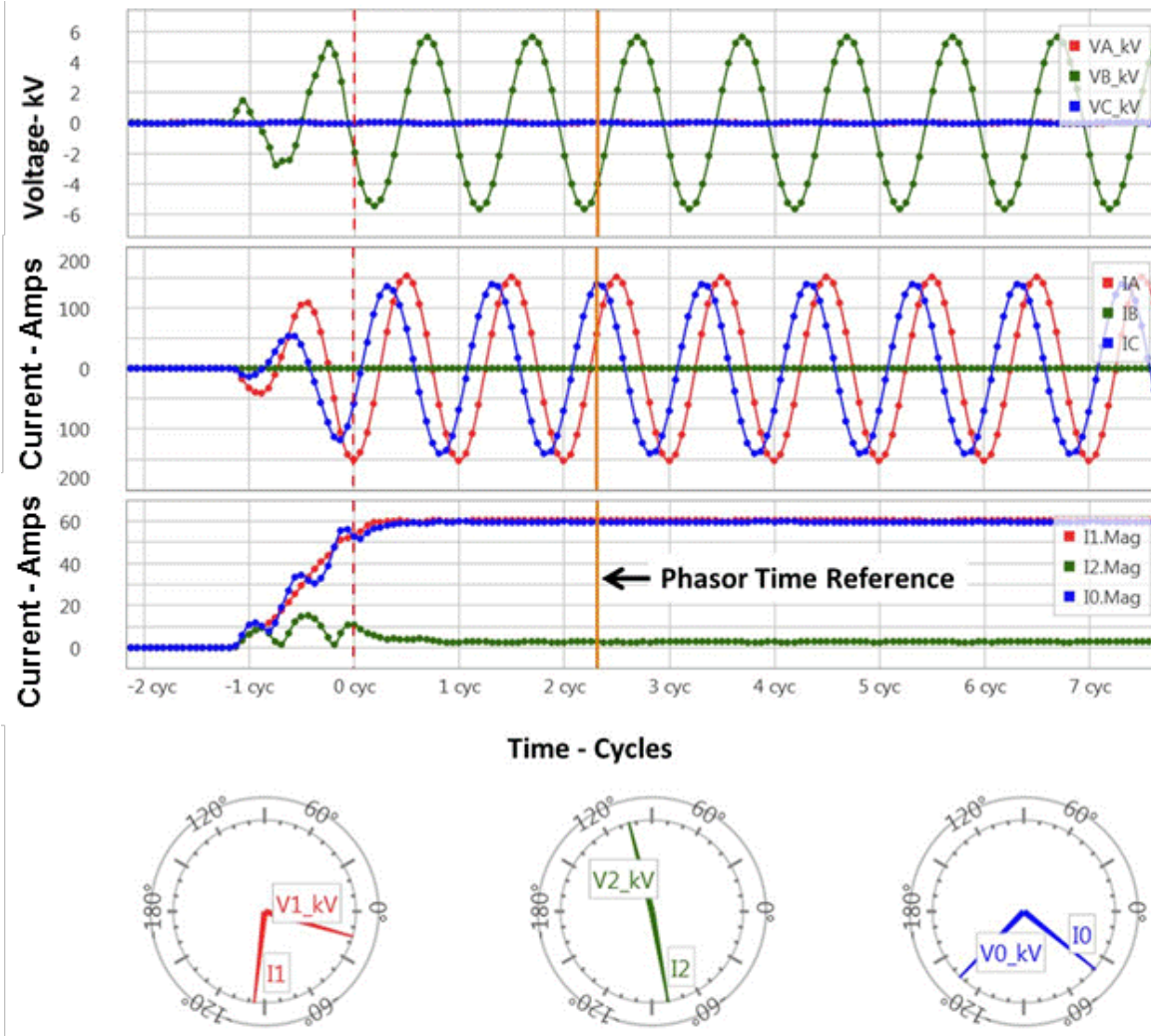


Figure 32: Inverter system response to line-to-line-to-ground fault

Lessons Learned:

Short circuit current analysis of the recorded ground fault confirmed that the negative sequence source impedance of the inverter source can be significantly high. Thus, negative sequence current can be too small to use in reliable relaying decisions. Inverter systems are typically ungrounded but they typically connect through an interconnection transformer that provides effective grounding to the high voltage network. As seen in previous

examples, the reliability of zero sequence relaying largely remains unhindered by high penetration of inverter sources as long as these sources are connected using a transformer which provides a path for zero sequence current.

4.2.2 Example 2 – Simulated Event

This section presents a simulation study of an actual system demonstrating challenges when applying a negative sequence overcurrent relay to the fault current contribution from Type III and Type IV wind turbine generation facilities.

System Description:

The actual system interconnecting five wind generation facilities was the same as discussed earlier in Example 4 described in Section 4.1.4 with details in Reference [20].

Simulated Phase-to-Phase Analysis:

A permanent phase-to-phase fault, as shown in Figure 27, was simulated. It was applied at $t=1.0$ s on the line interconnecting the 200×1.5 MW wind turbine facility to the rest of the grid. A negative sequence inverse-time overcurrent relay, denoted by 51Q, is assumed, on the line terminal leaving the wind generation facility. The primary negative sequence overcurrent (I_2) pickup was set at 330 A which was about 0.6 pu of the facility rated output. The following three scenarios were simulated to illustrate challenges of applying negative sequence overcurrent protection on a line terminal connected to the IBR facility.

Scenario 1: The wind facility as shown in Figure 27 was replaced with a synchronous generators of equal (300 MVA) capacity

Scenario 2: The original facility comprising of 200×1.5 MW Type IV wind turbine generators

Scenario 3: The original facility comprising of 200×1.5 MW Type III wind turbine generators

In Scenario 2, Type IV wind turbine generator's negative sequence current control system was modelled with gain settings: $k=0, 2$ and 6 . Note that the gain setting $k=0, 2$ and 6 emulate very high (infinite), $j0.5$ pu, and $j0.17$ p.u, respectively, negative sequence reactance paths through the IBR.

Figure 33 compares the negative (I_2) and positive sequence (I_1) current measured by the relay for the simulated fault under different scenarios. It also shows the statuses of the negative sequence inverse time pickup element (51Q). As expected, the negative sequence current was the highest (1.3 pu) for synchronous generator and the lowest (0.11 pu) for Type IV wind turbine generators with gain setting $k=0$. With gain setting $k=2$, the measured negative sequence current was still below the pickup setting (0.6 pu) of 51Q. Thus, the relay did not assert with gain settings $k=0$ and 2 . With gain setting $k=6$, I_2 injection by wind turbine was higher than the relay pick up and the relay tripped.

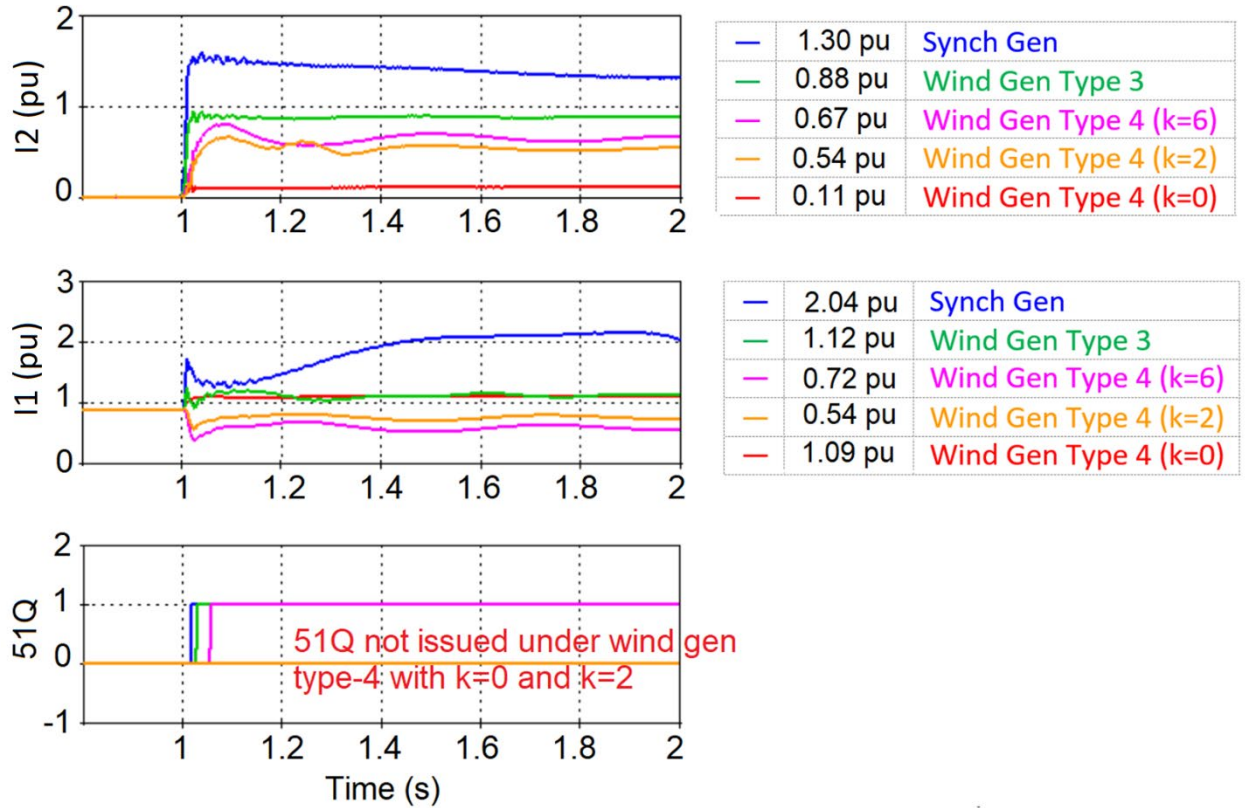


Figure 33: Negative and positive sequence currents along with statuses of 51Q for a phase-to-phase fault with different resources

Lessons Learned:

Depending upon the pick-up setting, this study illustrates that negative sequence overcurrent relaying poses a dependability risk for transmission lines connecting to Type IV wind generator facility whose control system has a low negative sequence current gain (k) injection setting. Type III generators supply some negative sequence current which is much smaller than the conventional synchronous generator for the same fault. Thus inverse-time coordination with the negative sequence overcurrent relays, where there is high penetration of Type III wind turbine generation, can be challenging without detailed understanding of each facility control system.

4.3 Phase Distance Relaying

Phase distance relaying is one of the most commonly used transmission line protection devices. Since reach and operating time of the distance relays are largely considered independent of short circuit current of the source behind relay, their application to the IBR connected lines was a normal choice until issues with their performance started to emerge under some IBR operating scenarios. If distance relay settings are determined based on traditional short circuit analysis using conventional source representations of IBRs, i.e. constant 60-Hz sources behind impedances and ignoring their controlled response

behavior, these settings are likely not reliable and may cause the relay to fail to trip or over-trip.

This section uses an actual example where misapplication of the phase distance relay from the post disturbance analysis was identified. It then identifies concerns and discusses solutions to help improve line protection reliability when applying phase distance relay operating on the IBR current contribution.

4.3.1 Misapplication of Phase Distance Relay

Following is an example where the line distance relay failed to operate on a fault internal to the line protection zone as well as could not provide backup for the external fault. The failure was largely attributed to a specific control mode setting of the IBR facility and the protection application engineers may not have been familiar with this operating mode.

System Description:

A Type IV wind resource facility is interconnected to the utility integrated system using a 7.5 km long 138 kV dedicated transmission line i.e. interconnection configuration similar to as shown between CB2 and CB4 in Figure 4. The facility consists of 34 wind turbine generators. Each wind turbine generator is rated at 3 MW. The facility is equipped with a STATCOM. This collector station of the facility connects through a 34.5/138 kV delta/wye-grounded transformer, with wye-grounded winding on the 138 kV side, to the interconnecting line. The interconnecting 138 kV line is protected by multifunction distance relays at CB2. Since the interconnecting line is a short line with a weak IBR source behind it, fast Zone 1 phase distance relaying was not used due to an anticipated high source-to-line impedance ratio [21]. Instead, a time-delayed Zone 2 distance relay was used.

Phase Fault Analysis:

On 29th July 2018, there was significant lightning activity in the area where the wind farm is located. Figure 34 is waveform capture by the distance relay for an external three-phase fault on that day. Once again the captured waveforms were processed by 60 Hz bandpass filter within the distance relay design and stored at a rate of 4 samples per cycle. In this figure: the top analog traces are phase currents; top middle analog traces are phase-to-ground voltages; middle analog traces are magnitudes of the three sequence currents; bottom middle analog traces are active and reactive power export from the wind turbine facility; bottom digital trace is status of Zone 2 distance relay element.

On the same day, the interconnecting 138 kV line also had a three-phase internal fault but the waveform records from that event were overwritten and are not available. However, the distance relays did not pick up during the internal three-phase fault and it was confirmed by the sequence of event recorder. In this application, the phase over current supervision was set too high at more than 420 A or 1.0 pu of total 100 MW plant capacity which is the maximum anticipated short current available from the facility. Even if the supervision element was set low-enough (at a small fraction of the plant capacity), the control response of the IBR from Figure 34 clearly illustrates that the distance relay had only a window of

opportunity of the first two cycles of three-phase fault inception to operate. During this time window, the current from the IBR facility remained at the pre-fault level. After two cycles, the output current from the facility dropped and eventually ceased. Even fast Zone 1 phase distance would have an inherent delay of up to one cycle (or more) so there would clearly be a race between relay operation and the IBR control system shutting down output current. Time delayed Zone 2 phase distance relays would have no chance to operate. Sequence of event data confirmed that the three-phase internal fault at CB2 was cleared by slow backup phase-to-phase undervoltage protection. This control is referred to as Zero Power Mode (ZPM) where the IBR facility enters into a temporary cease mode upon detecting a voltage depression.

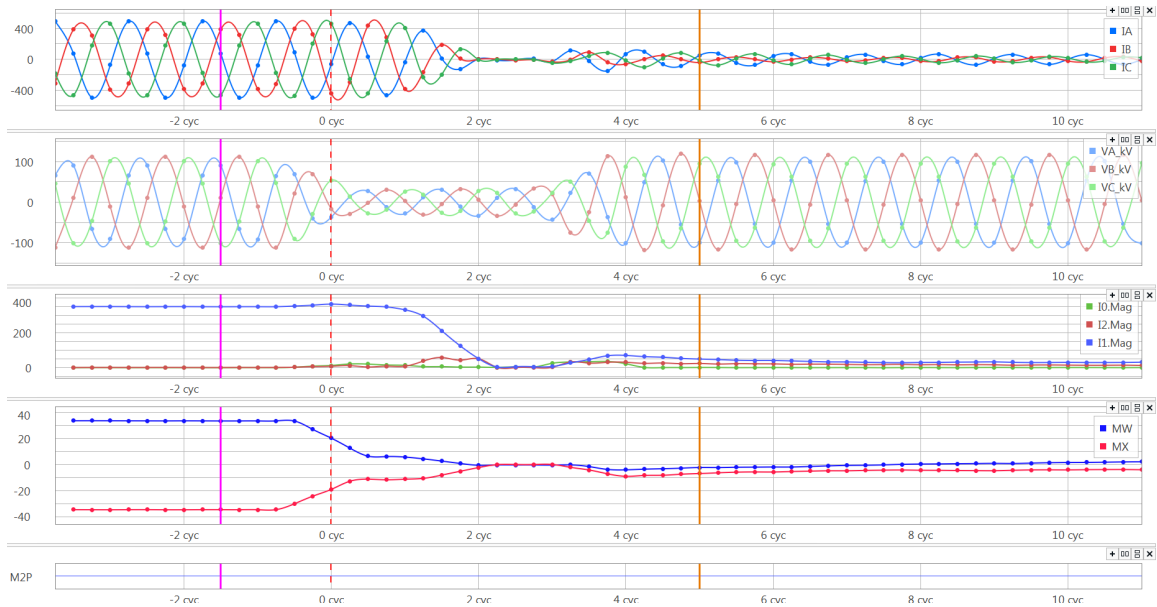


Figure 34: Three-phase short circuit response of a Type IV wind turbine generator operating in ZPM

Figure 35 shows waveforms captured by the relay for an external Phase A-to-B-to-ground fault on a 500 kV line in the region. This figure, unlike Figure 34, does not have power export and digital traces. The fault was cleared by the 500 kV line protection in about 3 cycles. Once again the positive sequence current from the wind resource facility remained at the pre-fault level for about two cycles into the fault then started to decrease before completely ceasing. The negative sequence current injection remained negligible during the fault. However, there was appreciable zero sequence current flow from the delta-wye transformer due to the unbalance voltage depression until the fault was cleared by the 500 kV line protection. The phase angle of zero sequence current, though not shown in Figure 35, was leading the corresponding sequence voltage by 90° during the external fault i.e. correctly indicating forward fault direction. Thus, the zero sequence overcurrent protection (with or without directionality) can be used for reliable line protection for ground faults.

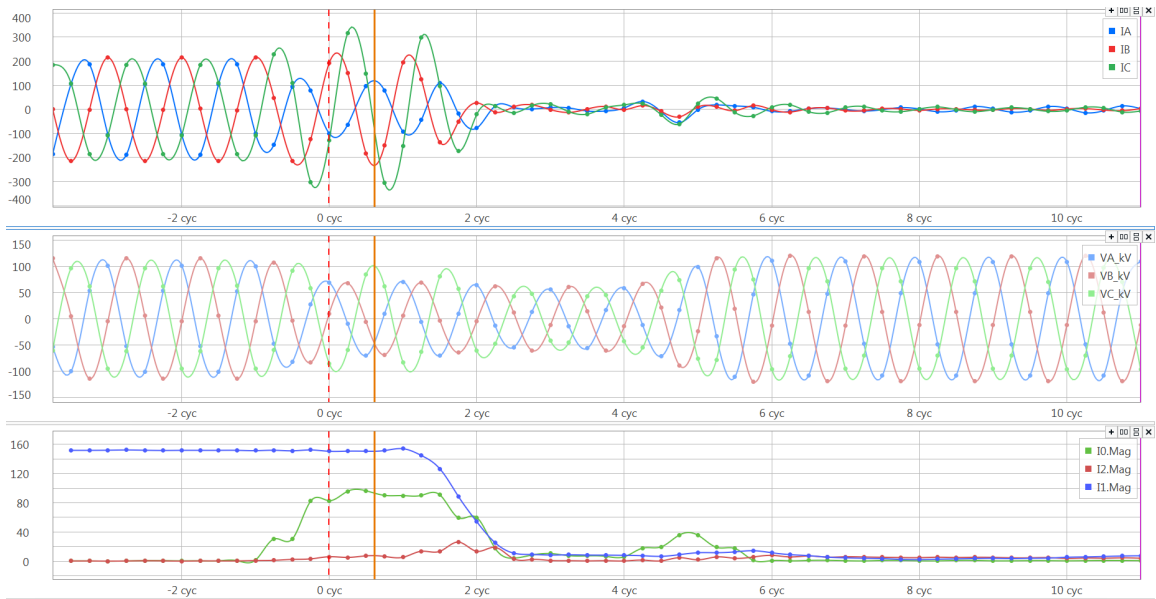


Figure 35: Data recorded by 138 kV line relay showing response of an IBR to an external unbalanced fault

Lesson Learned:

ZPM suspends the output current within two cycles of voltage depression when the IBR facility is still connected to the line. While this mode allows the IBR to ride through voltage depressions (low voltage ride through) during external faults, it also inhibits the distance (or overcurrent) protection for the internal line faults.

Prevalent deployment of ZPM operation in the existing IBR facilities and its risk to the system, particularly to traditional line protection, reliability was not well known until recently. On 16 August, 2016, the Blue Cut Fire in California caused a 500kV line fault. Though the fault was cleared in less than 3 cycles, approximately 1200 MW of solar generation was coincidentally either temporarily suspended or lost. Subsequent investigation by the NERC/WECC joint task force reported [22] that the majority of lost solar generation was caused by the zero operational mode i.e. IBR were configured to momentarily cease current injection for voltages outside their continuous operating range.

4.3.2 Countermeasures

Controlled short circuit response of the IBR facility leading to non-inductive and low magnitude fault contribution is a clear threat to reliability of traditional line distance or phase overcurrent protection schemes. These relaying schemes cannot be reliably applied when the facility is operating in ZPM. For the IBR facilities whose control system is designed to supply dynamic positive or negative sequence reactive current, additional countermeasures can be applied to attain an acceptable level of phase distance or overcurrent reliability.

Dynamically changing source impedance of IBR may pose a risk of over- or under-reach. However, this risk can't be assessed adequately without detailed three-phase analytic

model simulation of the IBR control system. In the absence of this detailed model simulation, some minimal precautions are essential to help reliability of the line distance relay with IBR. Distance relay phase supervision current is typically set to a low value (possibly to the minimum relay setting) to account for facility operation at low capacity. In electro-mechanical relays, low-set phase current supervision poses a risk of the relay operation on loss of potential. Modern multi-function relays offer sophisticated loss-of-potential logic that can avoid this security risk even with minimally set phase current supervision. Some new designs of distance relays use negative sequence current supervision. These relays cannot be reliably applied if the IBR control system suppresses negative sequence current during unbalanced faults.

It is important for backup protection to mitigate the dependability risk to the application of phase distance relaying. Backup protection comprised of a slow phase-to-phase under voltage protection for phase faults can be provided. Consequently, if the interconnecting line, such as a line between CB1 and CB3 in Figure 4 has tapped load, then the load gets subjected to poor power quality in the duration after opening of CB3 by the remote line protection until the slow undervoltage backup relay opens CB1 at the IBR terminal. To avoid subjecting the tapped load to poor power quality, a direct transfer trip (DTT) is utilized and initiated by the remote line protection at CB3 upon fault detection to the IBR line terminal to open CB1. Even with the DTT, slow undervoltage protection is still required as a local backup or last line of defense if the distance relay fails to operate (due to lack of current from IBR) while a DTT communication channel is down.

4.3.3 Backup Undervoltage Protection

The phase-to-phase undervoltage protection can be provided as backup to safeguard against risk of phase distance relay not picking up at the line terminal where IBR facilities are the short circuit source. In Figure 4, these locations are at CB1, CB2, CB5 (for contingency of CG2 not connected) and CB7 (for contingency of CG1 not connected). This protection is designed to override the slowest clearing faults on the adjacent circuits. In addition, the undervoltage protection should comply with interconnecting utility low voltage ride through requirement and other applicable regulatory standards. Using low voltage ride through of a large Canadian utility [23] and a newly approved North American Electric Reliability Corporation (NERC) standard PRC-024-01 [24], Figure 36 shows no-trip and permitted trip zones (set-points and time delays) used by the utility for back-up undervoltage protection.

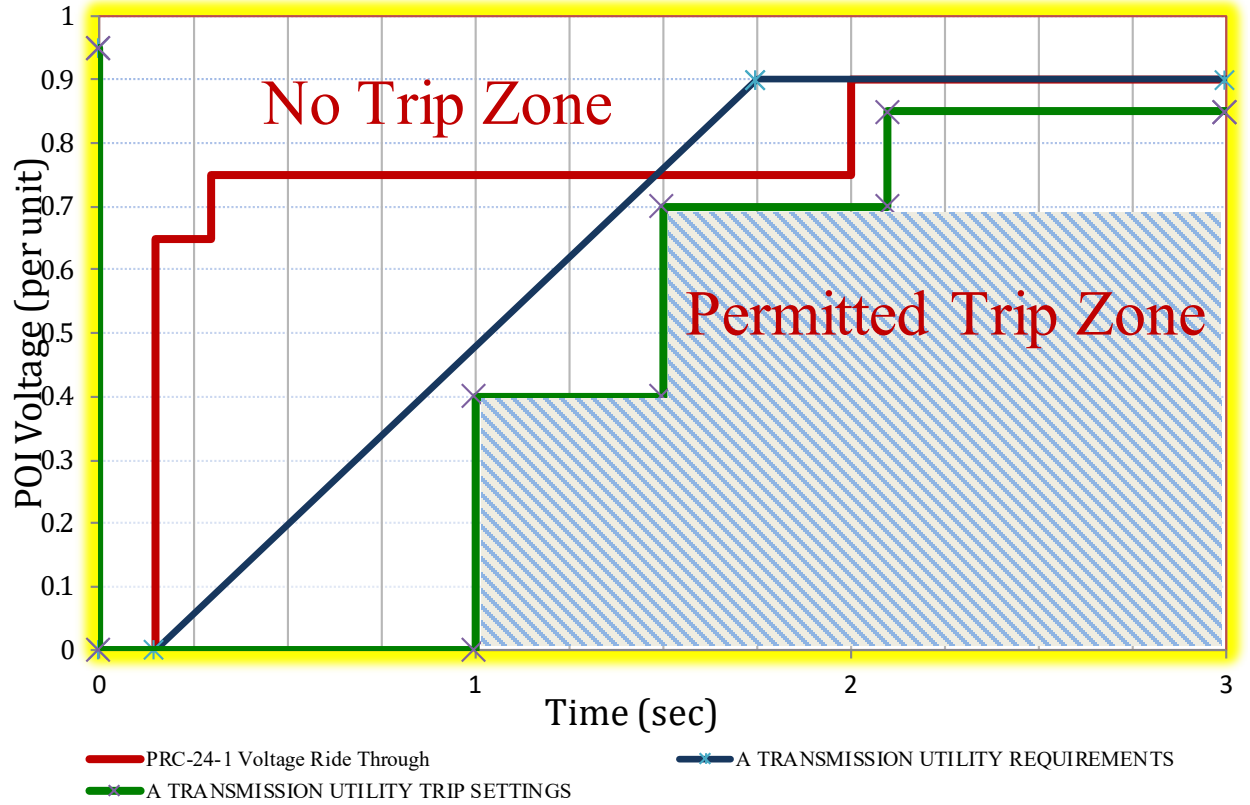


Figure 36: Phase-to-phase undervoltage protection trip zone

4.4 Permissive Overreaching Transfer Tripping Scheme

A permissive overreaching transfer trip (POTT) scheme provides high-speed fault clearing for the entire line that may be required for stability or power quality. The scheme requires only one signal exchange between two line terminals and thus communication bandwidth demand is minimal. It can also be applied in a transmission line serving tapped loads. With conventional sources with strong short circuit strength, a POTT scheme often uses directional mho phase distance relays for high-speed phase fault protection. Either directional ground distance or directional zero sequence overcurrent or directional negative sequence overcurrent relays for ground fault detection can be used for high speed ground fault detection. This section describes safeguards in the application of POTT scheme with IBR at one of the two terminals.

Referring to Figure 4, assume that there is a POTT scheme on the line between CB5 and CB8. This scheme requires some rudimentary safeguards to improve its reliability for contingency when the CG2 is out of service and the IBR facilities are the only sources behind the relay at CB5. During the CG2 contingency, the protection located at CB5 is expected to operate on short circuit current solely supplied by two IBR generating stations. As a result, it is important that the protection scheme includes provisions to manage the dependability risk of distance relays which are often used for the phase faults in the POTT scheme. The echo logic function [21] and DTT can manage this risk as long as conventional source (CG1) behind CB8 is strong which allows protection at that terminal to operate

reliably. Protection trips at CB8 involving echo logic, when phase distance relay at CB5 fails to operate for an internal line fault, are a few cycles slower than those trips without echo logic. Once the protection trips at CB8 (with or without echo logic), it also sends a direct transfer trip to open CB5. As discussed before, minimally set phase overcurrent supervision can be used in phase distance relays at both ends. As discussed earlier, slow undervoltage protection is required at CB5 as a local back-up or last line of defense if the distance relay fails to operate while communication link is down.

A zero sequence overcurrent fault detector coupled with a zero sequence voltage polarized directional relay in the POTT scheme can be used to avoid risk to ground fault protection. Zero sequence inverse-time overcurrent relaying, polarized by zero sequence voltage or current, can provide ground fault back-up if the communication channel is down.

A North American utility had experienced a line protection mis-operation when echo logic with the negative sequence directional polarization in an IBR installation was used. In this case, relays were installed to protect the line with configuration similar to the line between CB2 and CB4 in Figure 4. An external fault had occurred, as an example on the line between CB7 and CB10. The relay at CB4 incorrectly echoed back the permissive signal because it had failed to detect the reverse fault. The problem was attributed to the unreliable negative sequence current from the IBR contributing to the inability of the relay located at CB4 to detect the reverse fault. Following the misoperation, the echo logic was disabled. The option of zero sequence directional polarization was either not available or was not investigated.

4.5 Directional Comparison Blocking Scheme

A directional comparison blocking (DCB) scheme is a high-speed protection scheme which uses overreaching tripping elements and blocking elements. Multiphase faults are most often detected by phase distance elements, and ground faults are typically detected by either ground distance or ground overcurrent elements [21].

Referring to Figure 4, application of a DCB scheme on the line between CB5 and CB8 will work well when all generation sources are in service. However, if conventional generation source CG2 is out of service, then the line relays at CB5 may have difficulty in sensing faults on its line due to low levels of fault currents produced by the wind and solar farms. One possible way to handle this situation is to set the overcurrent supervision elements for the distance relays with extremely sensitivity so that they can sense faults even in the absence of CG2. Another possible solution for this situation is to employ a DTT with which the protective relays at CB8 send a DTT signal to trip CB5 when they trip CB8. Slow phase undervoltage and neutral overvoltage elements can serve as backup protection at CB5.

4.6 Line Current Differential Relaying

Digital teleprotection channels (with 64 kbps data rates) allow exchange of analog measurements over a long distance between line terminals by numerical line current differential relays. In spite of uncertainty of short circuit contributions from the IBR facilities, the differential relays can provide reliable phase and ground fault protection for

lines interconnecting IBR facilities to the utility grid. As long as the teleprotection channels are secure and available, the line protection will trip both line terminals simultaneously for all in zone faults and restrain for external faults [25]. At each terminal, local stepped phase distance and inverse-time zero sequence overcurrent directionalized by zero sequence polarization protections are included as backup for the contingency of teleprotection failure or breaker failure during the faults on the adjacent lines connected at the remote terminals. To compensate for the lack of reliable backup from phase distance relay operation at the line terminal with IBR, local back-up protections are provided. These are time-delayed phase-to-phase undervoltage protection and an independent direct transfer trip signal from the line terminal on the utility bus to the terminal connecting the interfaced source.

The line current differential as discussed provides fast, sensitive and reliable protection. The number of tapped loads that can be connected to the line is limited by the design of the differential relay i.e. ability of the relay to accept remote inputs via teleprotection channel. Even when the relay system has the capacity to accommodate tapped connections, there is higher cost to connect tapped loads due to added cost of digital teleprotection channels for remote measurements.

4.7 DTT to isolate generator for short circuits on utility system

When generation is being connected to the transmission system, it is important to consider the implication of islanding to minimize adverse impacts on loads. If a generator can be islanded, additional protection is required to prevent this from happening. Depending on the result of protection study, either a DTT from the remote utility terminal to POI and/or additional local relays monitoring voltage and frequency at POI are required. The additional over/undervoltage and over/underfrequency relays are applied as necessary to protect the load from damage. These relays are intended to trip the generator for the large voltage and frequency deviation that would tend to occur during a ‘local’ island condition.

Generally, one might consider using DTT as the primary protection to isolate IBRs for any fault on the utility system to prevent an islanding condition. Implementations of fiber optic and digital microwave for transfer trip communication paths are considered more secure and reliable than power line carrier or leased telephone circuits that are becoming obsolete. Since DTT applied over power line carrier trips for any loss of guard, field personnel typically need to conduct significant testing for all “carrier events”. Channel (non-licensed) radio circuits may not be considered reliable enough to use as a transfer trip communication medium. Satellite is reliable but is too slow to be used for DTT due to propagation delay.

4.8 Single-phase Trip and Reclose

A single-phase trip and automatic reclose scheme is sometimes used to enhance interconnection reliability when there is remote generation connected to the grid through a long radial extra-high voltage transmission line. The scheme maintains synchronous connection between the remote generation and grid throughout the single-phase open period. As a result, the connection between grid and generation is not lost during the most commonly encountered form of extra-high voltage transmission faults. The application of single-phase tripping and reclosing may not be viable when the IBR is unable to supply

negative sequence current. Before applying the scheme, the supplier can be consulted on the IBR's ability to supply the unbalanced load current without posing any risk to the transmission system during a single-phase open period.

5. Challenges to System Protection Schemes and Solutions

Unlike conventional rotating generators, solar and Type IV wind turbine generators do not have any inherent inertia. An increased footprint of IBR within a region significantly reduces the regional inertia that challenges reliability of the existing power swing blocking or out-of-step protection systems. The control system of a radially connected IBR (particularly Type III wind turbine generator) facility to a series capacitor compensated line can excite undamped and damaging subsynchronous oscillations. Unintentional IBR islanding can subject the customer loads in the island to poor power quality with damaging consequences. This section discusses these new challenges and methods to address them.

5.1 Power Swing Protection Schemes

The impact of the connection of IBR on the performance of power swing protection is investigated in this section. Since IBR have no inherent rotational inertia, power swing characteristics may be substantially different than that under synchronous generation. Such different swing characteristics may result in the misoperation of a legacy power swing protection system. This section provides examples of the misoperation of power swing blocking (PSB) and Out-of-step Tripping (OST) functions of power swing protection. This simulation based study was performed on a modified version of the IEEE PSRC D6 working group [26] test system that includes wind generation.

System Description:

Figure 37 shows the test system of this study. Details of the system parameters can be found in Reference [26]. A fault on Line 1 (L1) or Line 2 (L2) followed by a loss of the faulted line resulted in a power swing which can be stable or unstable depending on fault clearing time and pre-fault loading conditions. The power swing was between cluster of generators (G1 to G4) and the rest of the system. A multifunction relay denoted by R21 incorporating distance and power swing protection was added on Substation A. The power swing relay uses a mho-with-blinders detection scheme which employs three concentric circles restrained by three sets of blinders to maximize the chances that the minimum load impedance (during maximum power flow through the line) seen by the relay R21 is well outside of the outer blinder. The PSB function uses the outer and the middle characteristics along with the PSB time delay (e.g. three cycles) to detect a power swing. The middle characteristic encompasses all protection tripping zone (Zones 1, 2 and 3) of the distance elements to supervise their tripping. As soon as the swing impedance trajectory enters the outer characteristics, a timer counts down the PSB time delay. If the timer expires before the swing impedance crosses the middle element, the event is declared as a power swing and a PSB signal is issued to block the zones of distance protection. However, if the swing impedance crosses the middle element before the timer expires, the event is classified as a fault and PSB is not issued to allow the distance relay to pick up the fault. The OST function uses the inner characteristics which are designed such that the impedance trajectory of the

most severe stable swing does not enter it. Hence, if the impedance trajectory of a swing crosses the inner element, the swing is declared unstable. Otherwise, the swing is classified as stable.

The simulation study illustrated that the power swing relay successfully detected stable/unstable power swings under synchronous generation scenario. However, misoperations were identified under wind generation.

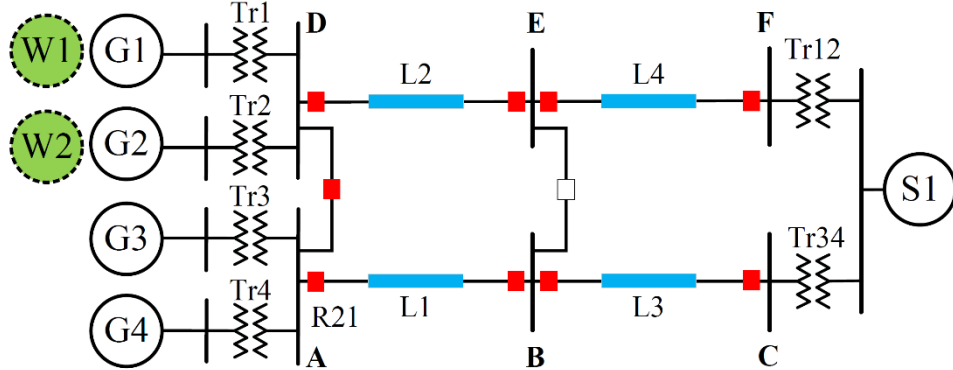


Figure 37: IEEE PSRC WG-D6 test system including wind generation

PSB Misoperation due to IBR

A three-phase fault with a time duration of 270 ms was applied on line L2 close to Substation E followed by the outage of lines L2 and L4 at $t = 2$ s resulting in an out-of-step condition. The event was simulated under 0% and 25% IBR penetration levels realized by replacing Generator G1 with a Type III or Type IV wind turbine or solar PV generator (W1). However results from simulation of Type III wind turbine generator are presented in this section because they were similar for Type IV wind turbine and solar PV generator, assuming the same inverter controls.

Figure 38 depicts the swing impedance trajectory during the first 30 cycles of the power swing, and Figure 39 shows the response of distance and power swing elements. As shown under 0% wind turbine generation, the impedance trajectory crossed the outer and middle elements in more than the PSB time delay setting of three cycles (48 ms). Hence, the PSB successfully detected the swing and issued a PSB signal to block all tripping zones of distance protection elements. Under 25% wind turbine generation, the impedance trajectory crossed the outer and middle elements in less than the PSB time delay setting. Hence, the power swing protection failed to detect the power swing and did not issue the PSB signal. As the swing continued, the impedance trajectory entered Zone 2, and the power swing was mistakenly declared as a fault in Zone 2 which was picked up by the distance protection. This could result in undesired tripping of line L1.

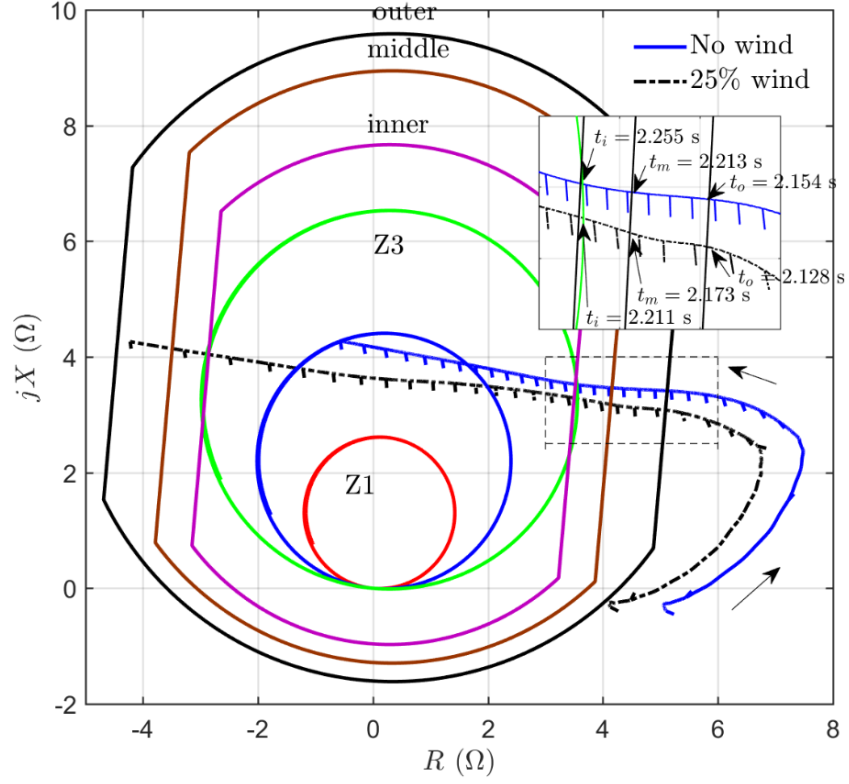


Figure 38: Swing impedance trajectory under 0% (blue) and 25% wind generation (dashed black)

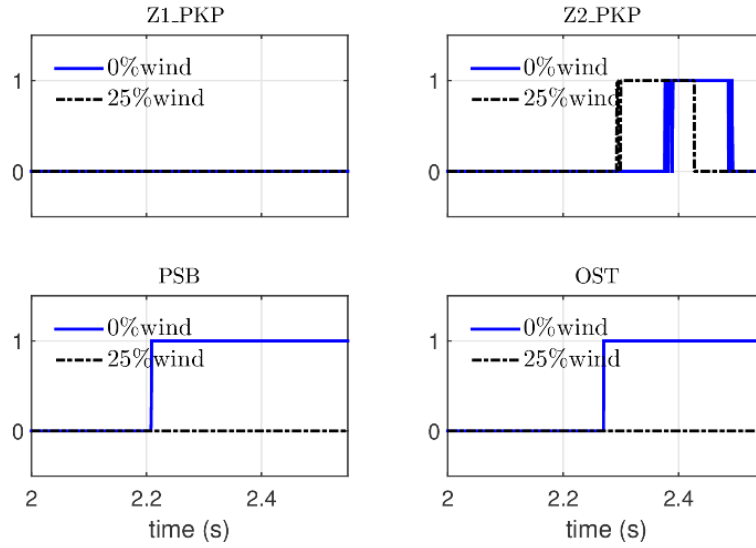


Figure 39: Distance and power swing relay signals under 0% (blue) and 25% wind generation (dashed black)

Note that distance relay zone 1 pickup is Z1_PKP; distance relay zone 2 pickup is Z2_PKP; power swing PSB signal is PSB; and power swing OST signal is OST.

OST Misoperation due to IBR

This case study compares the most severe stable swing under 0% and 50% IBR levels. In simulations, Synchronous Generators G1 and G2 were replaced with Type III or Type IV wind turbine or solar PV generators (W1 and W2). The simulations shown in this section were conducted with Type III wind turbine generator models, however similar results were observed with Type IV wind turbine and solar PV generators, assuming the same inverter controls.

Under 0% wind, the most severe stable swing was caused by an 88 ms three-phase fault on line L2 close to substation E followed by the outage of lines L2 and L4. But under 50% wind generation, the most severe stable swing was caused by 120 ms fault. Figure 40 depicts the swing impedance trajectory of the most severe stable swing under 0% and 50% wind generation, and Figure 41 shows the response of power swing protection.

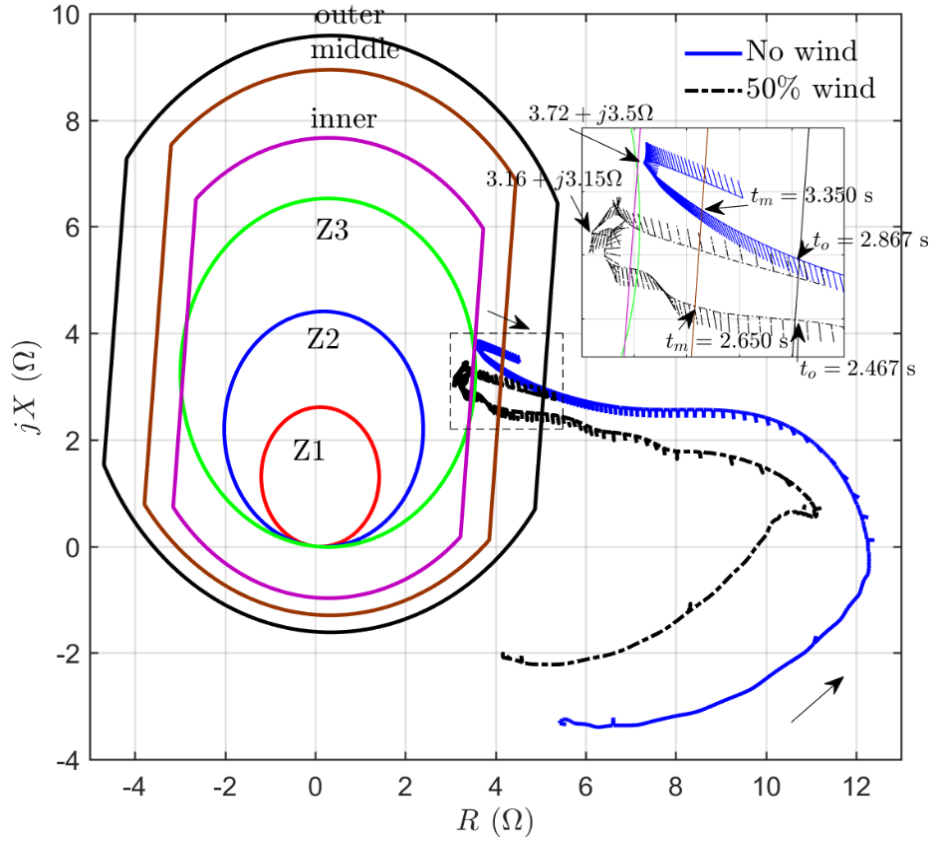


Figure 40: Swing impedance trajectory of the most severe stable swing under 0% (blue) and 50% wind generation (dashed black)

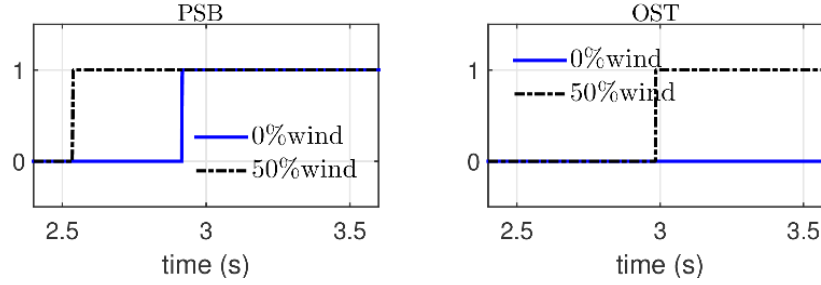


Figure 41: Power swing relay signals under 0% (blue) and 50% wind generation (dashed black)

Note that power swing PSB signal is PSB and power swing OST signal is OST.

As shown, under 0% wind generation the impedance trajectory reversed direction outside the inner element. Hence, the power swing protection declared the swing as stable, issued a PSB signal, and did not issue an OST signal. However under 50% wind generation, the impedance trajectory reversed direction inside the inner element. Thus, the relay mistakenly declared an OOS condition and issued an OST signal.

Note that the OST function may have a timer. However, the IBR-related OST misoperation is not associated with the setting of the timer, since it is due to the resistive reach of the inner blinder.

Lessons Learned

Case studies have presented examples of PSB and OST misoperation under high levels of IBR. The examples above show that swing characteristics are impacted significantly by Type III wind generation. In particular, conducted simulation showed that IBR affect both the rate of change of the swing impedance and the swing trajectory and, hence, may impact the operation of both PSB and OST.

The PSB misoperation scenario illustrates that high level of IBR integration may increase the rate of change of the swing impedance vector and cause PSB misoperation. To avoid such misoperation, potential solutions could be to reduce the PSB time delay or set the middle blinder closer to the inner blinder or a combination of both. Nevertheless, in general special care needs to be taken when reducing the PSB time delay since a very small PSB time delay may cause the PSB function to misinterpret a fault for a power swing and unintentionally block distance protection elements during a fault. In addition, the middle blinder should encompass the largest distance protection zone to be blocked by PSB. Reducing the resistive reach of this blinder may create the risk of zone encroachment.

The OST misoperation scenario illustrates that the swing trajectory might change in the presence of IBRs and the inner blinder setting should be adjusted accordingly. Placing the inner impedance measurement element outside the largest distance protection region that is to be blocked will allow enough time to carry out blocking of the distance elements after a swing is detected.. Setting the inner blinder closer to the line impedance may create an issue if the reduced blinder encroaches on a supervised protection zone. This challenge

may be addressed by using a quadrilateral or other characteristic to reduce the resistive reach of the impacted protection zone.

Impact of IBRs on power swing relaying is being investigated in more detail by another IEEE PSRC Working Group (D29).

5.2 Synchronous Condenser Application

Solar and wind generators are displacing retiring fossil fuel and nuclear generation in many transmission jurisdictions where they are now becoming predominant resources. Unlike conventional generators, IBR do not have stored kinetic energy as inertia and have limited ability to supply reactive support during voltage depressions. The reduction in inertia causes the system frequency to vary more rapidly during load changes and makes maintaining angular stability more challenging for the remaining conventional generators. Likewise, the reduction in reactive power support presents difficulty in keeping system voltage at the rated level during external faults and makes maintaining voltage stability more challenging.

One North American utility in Southern California deployed synchronous condensers to tackle the detrimental effects posed by displacement of conventional generators with IBR [27]. Other utilities around the world are employing synchronous condensers [28]. Similar to conventional generators, synchronous condensers have an inherent inertia response to system events because they have electromagnetic coupling to the grid. This inertia helps to keep the system frequency stable by releasing stored kinetic energy when the system frequency falls and absorbing energy when the system frequency increases. Synchronous condensers are also equipped with an excitation system; providing the ability to supply or absorb reactive power. This helps to keep the system voltage stable. During the Blue Cut Fire event on 16 August 2016 where 1200 MW solar generation was lost in California [22], data recordings showed that these synchronous condensers supported the system by supplying active and reactive power proportional to their ratings which helped to compensate the impacts of some lost solar resources.

The short circuit current responses of a synchronous condenser and a conventional generator are comparable immediately following the short circuit initiation. A synchronous condenser can supply sub-transient short circuit current from three to six times of its rating during balanced faults. Similarly, they can typically supply enough negative sequence current (in the proper angular relationship) during unbalanced system faults to eliminate protection reliability concerns.

Synchronous condensers, operating in parallel with IBRs, can help resolve protection issues. However, their installation requires capital investments and on-going operational expenses. Thus, synchronous condenser applications may be limited and considered practical solutions only alongside large scale utility IBR installations.

5.3 Interaction of IBR with Series Compensated Transmission Line

Different from traditional subsynchronous resonance (SSR) between a synchronous generator turbine shaft and a series compensated transmission line, the control system of

IBR, particularly Type III wind generation, can interact with a series compensated transmission line to create a new type of subsynchronous oscillation (SSO) phenomena, which is often specifically categorized as subsynchronous control interaction (SSCI). The 2009 SSCI event on the AEP system in Texas awoke the industry to this possibility [29]. Despite the efforts that have been put into system planning studies to minimize the risk of SSCI when the system is subjected to possible contingencies and wind turbines are required to be equipped with SSCI mitigation mechanism, three reported SSCI events in 2017 on the AEP transmission system within ERCOT emphasize the challenges of effectively combating SSCI problem [30]. All three reported SSCI events on the AEP system started after wind farms were radially connected to series compensated transmission lines resulting from adjacent transmission line outages.

System Description:

Figure 42 illustrates the transmission network configuration of two 2017 SSCI events on the AEP transmission system.

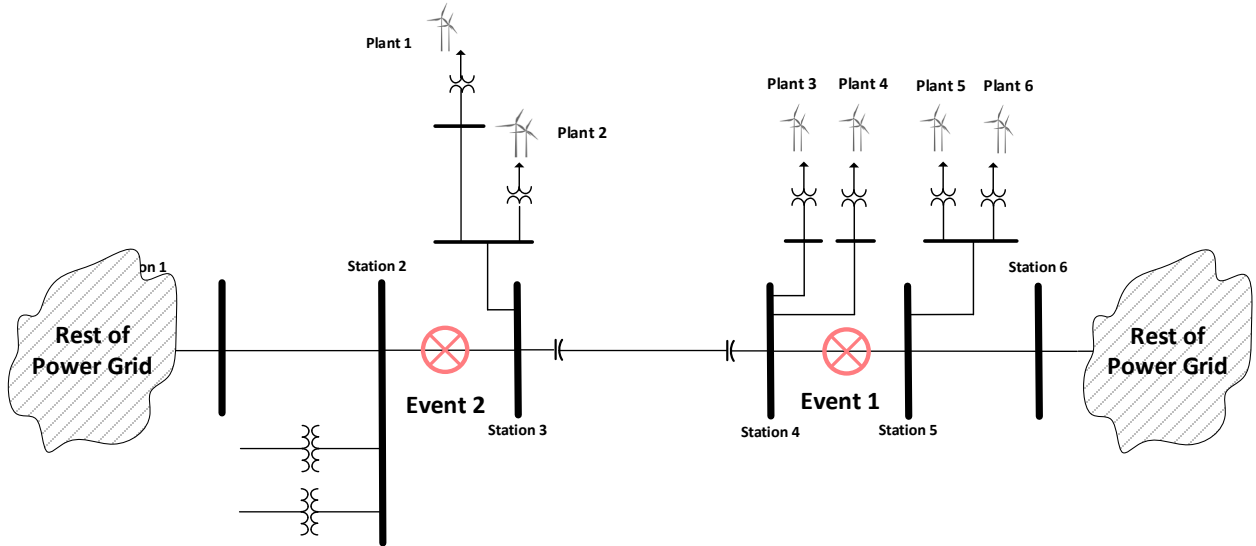


Figure 42: Transmission network configuration of AEP 2017 SSCI events

Event Analysis

In event 1, Plant 3 and Plant 4 were radially connected to the rest of transmission system through the 345kV series compensated transmission line between Station 3 and Station 4 after line between Station 4 and Station 5 was tripped. Figure 43 shows the line voltage and current recorded by line relay at Station 4 during the oscillation. Spectrum analysis result of the line current during oscillation reveals subsynchronous oscillation frequency in this event is around 25.6 Hz and the magnitude of the subsynchronous frequency component is almost twice as much as the fundamental frequency (60Hz) component. The oscillation stopped after Plant 3 and Plant 4 were tripped off the transmission system by the plant protection systems.

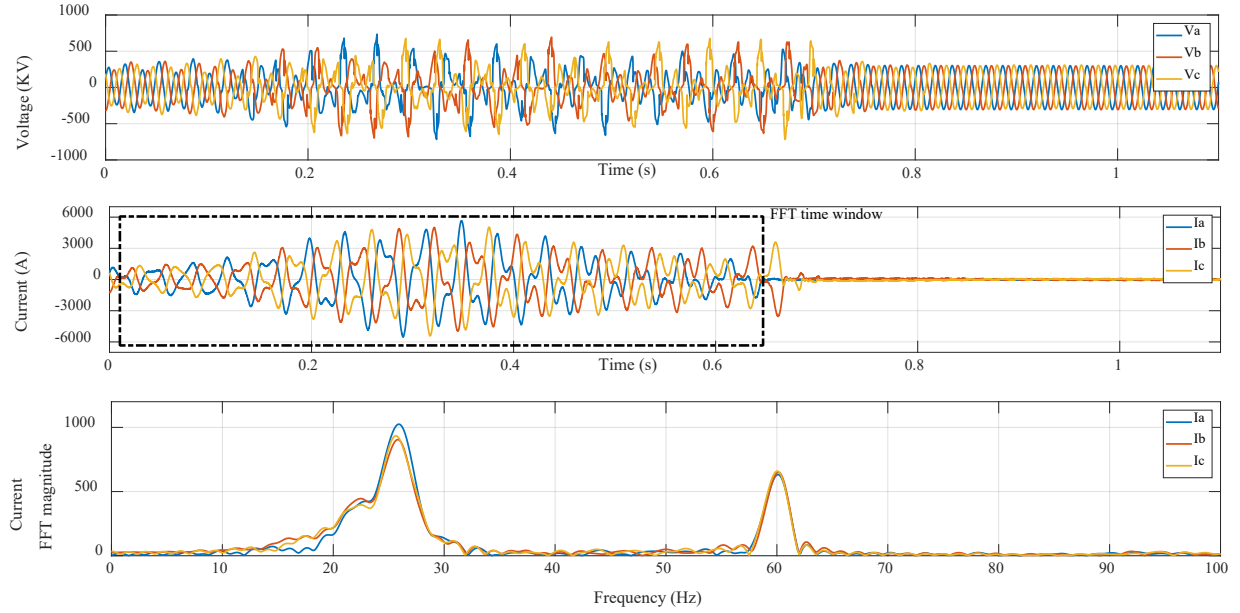


Figure 43: Event 1 line voltage and current on Station 4- Station 3 line at Station 4 and line current frequency spectrum

Similarly, in event 2, the oscillation started after the transmission line between Station 2 and Station 3 was tripped due to a line fault that resulted in Plant 1 and Plant 2 being radially connected to the rest of the transmission system. Figure 44 shows the line voltage and current recorded by line relay at Station 3 during the oscillation. Spectrum analysis result of the line current during oscillation reveals subsynchronous oscillation frequency in this event is around 22.5 Hz. The oscillation grew significantly in 400 ms after the line tripping. The oscillation stopped after the wind farm transformer differential protection inadvertently tripped Plant 1 and Plant 2 off of the grid due to the excessive subsynchronous harmonics in both voltages and currents.

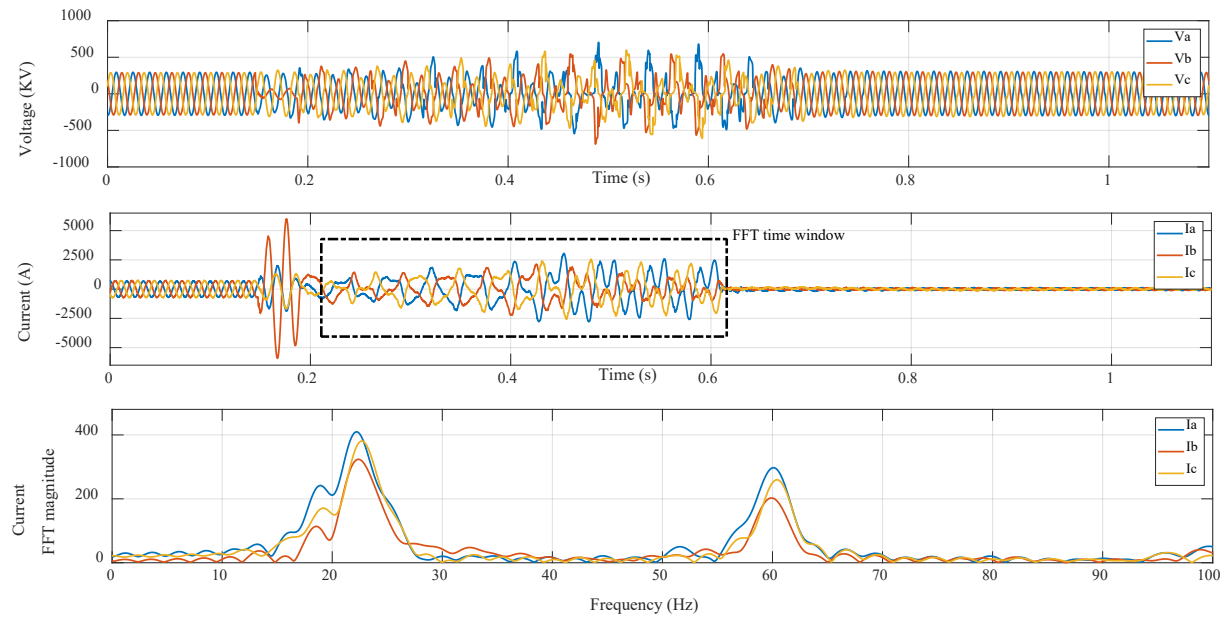


Figure 44: Event 2 line voltage and current on Station 4- Station 3 line at Station 3 and line current frequency spectrum

The fact that Plant 1, Plant 2, Plant 3 and Plant 4 turbines were from different manufacturers reminds us that the SSCI issue is beyond individual wind turbine manufacturer design issue. SSCI risks in the area were actually identified at planning stage through extensive dynamic simulations and SSCI mitigations were required to be implemented in the turbine control systems by manufacturers. Given the efforts that have been put into preventing SSCI from occurring by regional transmission planning coordinator, transmission owners, generation owners and equipment manufacturers, the occurrence of these events show SSCI remains as a challenge to transmission system planning, protection, and operations.

Lessons Learned

The undesired impacts of SSCI on power system equipment and system operation include:

- The quick rising voltage and current magnitude could damage system primary equipment including series capacitor banks, synchronous generator turbine shafts, power transformers, etc. Because most relays operate on fundamental frequency quantities, conventional relays may not react fast enough to current and voltages with significant subsynchronous component and, therefore, to protect power system equipment.
- Protective relaying, such as transformer protection, which may misoperate in the presence of excessive subsynchronous harmonic components. In fact, the misoperation of the transformer differential in event 2 exemplifies this.
- Although SSCI is a relative local power system phenomenon, delayed clearing or mitigating of the oscillation could leave the oscillation to propagate through the power system, affect neighboring generation units including inverter based and traditional

synchronous power generation units, and create undesired cascading effect to power system.

In addition, to continue to improve the wind turbine dynamic models and verify the effectiveness of SSCI mitigation functions through simulations, protection measures have been defined to detect subsynchronous oscillation in a timely fashion and to automatically take proper measures to mitigate the oscillation if SSO condition occurs [31]. Some key protection function requirements are suggested to maximize the reliability of SSO protection:

- Protection function takes three-phase voltages and three-phase currents and allows user to set different thresholds of voltage and current. These SSCI events demonstrated that voltage and current have different oscillation magnitudes during the event. It is important to be able to have voltage and current oscillation detection operate independently.
- Oscillation protection function allows the user to specify the upper and lower cutoff frequency of the bandpass filter that is designed to extract the subsynchronous component from the measurement. For example, for a 60 Hz power system, the limit of the lower cutoff frequency (f_{low}) can be **10 Hz** and the limit of the upper cutoff frequency (f_{upper}) can be **50 Hz**. The stop band attenuation of the band pass filter is at least 40 dB to prevent any other system transients to encroach the SSO protection function.
- Oscillation protection function may be considered to detect the oscillation with the maximum time delay (T_{delay}) of $2.5/f_{osc}$ second, where $f_{osc} = \frac{f_{low} + f_{upper}}{2}$. For example, for a subsynchronous oscillation with f_{osc} as 25 Hz, the oscillation protection function is expected to detect the oscillation within 100 ms. Fast and reliable SSCI detection is important because past events have demonstrated that oscillation voltages can grow into a level that can be damaging to the power system elements within a few hundred milliseconds.
- Oscillation protection function may be considered to provide multiple levels of oscillation detection to allow early warnings at low level oscillations and fast action on strong oscillations.

With fast, reliable, and secure SSO protection, a SSO relay can be configured to automatically bypass a series capacitor bank and possibly trip wind farms off the grid to disconnect the source of SSCI and protect equipment from damages resulting from SSCI conditions.

5.4 Solution to Control Interactions

SSCI is a phenomenon which commonly occurs due to IBR controller interaction with radially connected fixed series compensated systems [32][33][34][35][36]. In addition to those discussed in the previous section, there are several real-world events that have taken place due to SSCI issues [32][33].

To mitigate and/or protect the system from the SSCI issues associated with IBRs, much research and development has been attempted, specifically by redesigning the existing IBR controllers to provide sufficient damping at the subsynchronous frequency range. Although the modern controllers in IBRs have been developed with the capability to mitigate the SSCI conditions, in some specific applications (with Type III turbines), it may not be possible to eliminate the SSCI phenomenon due to the dynamic nature of the controller interactions and continuously changing topological network conditions. In addition, there are several existing installations that are equipped with older controller designs which are also susceptible to SSCI conditions. Therefore, the use of an appropriate SSCI mitigation method to prevent system damages against such SSCI conditions is essential.

During recent years, the applicability of various protection/mitigation against such conditions has become attractive. Based on the reported practical application case studies, SSCI mitigation methods used by different utilities can be broadly categorized into two main types (i) Topology based mitigation and (ii) Mitigation using a subsynchronous resonance (SSR) relay.

In this section, a series compensated transmission system connected to a wind-based IBR has been considered as an example case study to describe possible approaches to mitigate the SSCI conditions.

5.4.1 Topology Based Mitigation

Based on the field reported events, and detailed simulation studies, it has been proven that the topology plays a major role in SSCI phenomenon. Thus, the applicability of topology-based mitigation has been widely considered in many applications. Identification of topologies that leads to SSCI may sometimes need detailed system studies. These studies may include a combination of eigenvalue analysis, frequency scans and electromagnetic transient simulations.

For the case study considered, bypassing the series capacitor or tripping the wind generation can be considered as possible solutions; however, it is questionable if the system studies sufficiently cover all possible scenarios leading to SSCI conditions.

5.4.2 Mitigation Using SSR Relay

Figure 45 shows the arrangement of a SSR relay configured to bypass the series capacitors. In this arrangement, the SSR relay takes the voltage and current measurements from the transmission line for analysis. For this application, measurements can be taken at POI on the transmission line and provide a more economical advantage for the users compared to the second approach explained below. Determination of protection settings for this application may require the analysis of multiple contingencies as decisions are made based on the full current flow on the line. It should be noted that in this arrangement, the relay can also be used to trip the transmission lines circuit breakers instead of bypassing the capacitors.

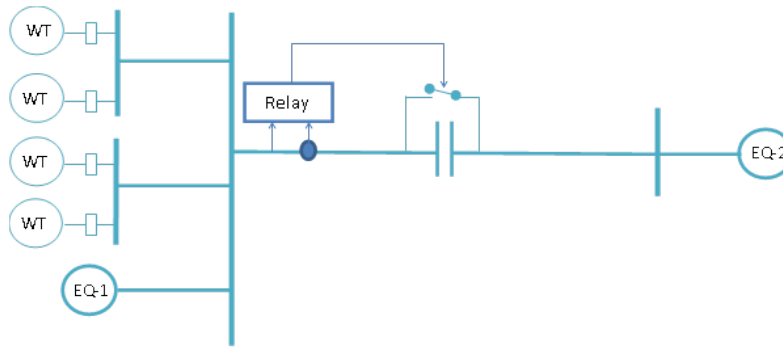


Figure 45: Relay arrangement for bypassing the series capacitor

Figure 46 shows the arrangement of the SSR relays configured to trip the windfarms to avoid the SSCI phenomenon. In this arrangement, the relays are installed at wind farm collector locations. Depending on the arrangement of the collector feeders, measurements from multiple or single points may be required. It is important to assess and appropriately select such requirements. However, the use of local measurements provides more flexibility and selectivity for settings compared to the approach explained above. It should be noted that in this arrangement, decisions from relays can also be used to by-pass the series capacitors or trip the transmission line completely.

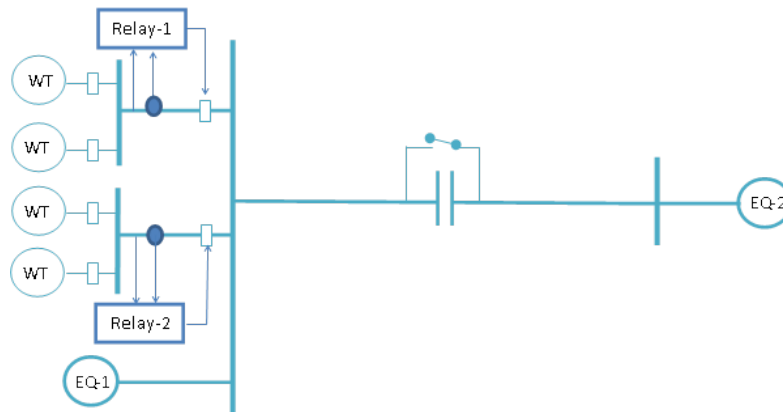


Figure 46: Relay arrangement for tripping the windfarm

5.5 IBR Islanding Considerations

Anti-islanding schemes continue to evolve with technology and algorithm advancements within inverter controls. Historically the transmission owner was responsible for disconnecting non-utility generation connected to transmission system during an islanding event. Modern inverters may have built-in active and passive anti-islanding controls that will shut down the source in an islanding event. As with most protection challenges selecting the best anti-islanding scheme requires a delicate balance between maintaining

acceptable levels of power quality, maintaining minimum voltage and frequency ride through support during transient events, and cost.

5.5.1 Anti-Islanding Guides and NERC Standards

Inverters may have built-in passive and/or active anti-islanding detection systems. Passive anti-islanding detection methods are based on a (passive) measurement of the grid values. Active method means that the inverter makes something active and gets back a reaction of the grid. This reaction is very different in case of islanding grid, compared with normal grid operation. Inverters may use voltage and frequency measurements as passive islanding methods, and employ the escalating frequency shift method as an active detection system.

IEEE 1547-2018 [16]:

The IEEE standard 1547-2018 applies mainly to distribution connected resources. For an unintentional island in which the distribution resource energizes a portion of the Area Electric Power System (EPS) through POI, the distribution resource shall detect the island, cease to energize the Area EPS, and trip within 2 seconds of the formation of an island. It is important that the anti-islanding scheme is secure enough to not operate under false conditions.

NERC Reliability Guideline - BPS-Connected Inverter-Based Resource Performance, September 2018 [37]:

As a result of the ongoing analyses of large bulk power system (BPS) disturbances in the Western Interconnection involving inverter-based resources, particularly solar PV resources, NERC formed the Inverter-Based Resource Performance Task Force (IRPTF) to develop recommended performance specifications for BPS-connected IBRs during steady-state and dynamic system conditions. The Blue Cut Fire and Canyon 2 Fire events in the Southern California area led to key disturbance reports and NERC alerts to identify the extent of conditions and develop recommended mitigating actions [22].

This guideline focuses on IBRs directly connected to the BPS. While NERC Reliability Standards only apply to BES resources, this guideline is also relevant to smaller IBRs that are still connected to the BPS. The guideline does not cover resources connected to the distribution system and instead recommends the use of IEEE std. 1547-2018 for these resources.

On 16th August, 2016, the Blue Cut Fire caused a 500kV line fault that cleared normally (less than 3 cycles) but this fault also resulted in the widespread loss of about 1200 MW of solar resources. This event was the primary focus of the NERC/WECC joint task force, which was formed prior to the NERC Inverter-Based Resource Performance Task Force. The task force published a disturbance report that identified the following key findings:

- Inverters were tripping erroneously on instantaneous frequency measurements
- The majority of inverters are configured to momentarily cease injection of current for voltages outside the continuous operating range around 0.9 – 1.1 pu

The guideline recommends disabling the following internal inverter protection functions:

- Unintentional islanding protection
- Passive anti-islanding protection

Most BPS connected IBRs are required to provide support and ride-through voltage and frequency grid disturbances. These requirements complicate inverter's internal anti-islanding protection schemes that look at changes in voltages and frequencies. If anti-islanding protection is necessary, transfer trip based anti-islanding scheme remains a viable option for application.

5.5.2 Utility Anti-Islanding Philosophy Example

Most utilities do not allow sustainable islands for N-1 contingencies because the generator, regardless of its type and size, may not be able to hold voltages and frequencies within acceptable limits, posing a threat to utility and customer equipment. The following presents an analysis approach used by a utility to determine if anti-islanding protection is necessary. Note that this methodology is used for analyzing interconnection of solar PV based resources only.

1. Are there less than three transmission lines at POI that the solar site is interconnecting to (not including radial substations)? If yes anti-islanding protection may be required, if no then scheme is not required.
2. Determine the boundary of the study. Any radial transmission facilities from the POI are included in the study. Follow each transmission line out from POI and determine if the substation at the remote end has less than three transmission lines (including the transmission line from POI). If the substation has less than three transmission lines add the substation to the analysis and repeat this step for the 2nd transmission line at this substation, adding the substation to the analysis and repeating the step if less than three transmission lines exist. For the boundaries of the study note all transmission lines, substations, wholesale points-of-delivery and any transmission connected loads.
3. Determine the interconnecting generation site nameplate MW along with any interconnected generation within the boundary of the study.
4. Determine loads within the boundary of the study; include distribution load, wholesale point-of-deliveries, transmission load, and any radial loads. If the peak summer local load is less than the local interconnected generation anti-islanding is required and no further analysis is needed, otherwise determination of local minimum load is necessary. Various methods can be used to determine local minimum load including:

Method 1: Create a calculation based-on metered values (e.g. using metered data captured by the utility PI server) for these loads and examine what minimum loads exist between the hours of 10:00 and 16:00 during the shoulder load seasons (spring and fall).

Method 2: Determine what a reasonable minimum historic load for this group of loads is based upon historic PI line loading into the local area using the same time period as in Method 1.

Method 3: Determine what the peak summer load of the local area and scale the load to the predominant system minimum load during the time period of Method 1.

5. Calculate 66% of the minimum load determined in Step 4. Compare the generation value to the 66% shoulder minimum load value. If the generation value is greater than the load, an anti-islanding scheme is required. If the load is greater than the generation an anti-islanding scheme is not required.

6. Conclusions

This section lists reliability risks to protection systems operating in an area with large presence of IBRs and proposes countermeasures that can be used in the existing protection to overcome these risks.

- Dynamic variations in IBR apparent source impedance (magnitude and phase angle) may cause polarized mho relays to over- or under-reach significantly or low facility output current may be not sufficient to permit distance relay operation, particularly during adverse weather conditions. Backup undervoltage protection, located on the line terminal at the IBR facility, can be used. This protection, set to operate on phase-to-phase voltage and coordinated with low-voltage ride through requirements, provides back-up to phase distance protection.
- Operation of IBR facilities in ZPM poses system and line protection reliability risks and thus their usage in future transmission connected IBR facilities are generally discouraged. Traditional distance and phase overcurrent schemes cannot be reliably applied on the line connecting IBR facilities operating in ZPM. Remote DTT, backed up by slow local phase-to-phase undervoltage, can be used as phase fault protection in the existing facilities.
- IBRs response to faults could be split into two time periods; before IBR limits the fault current and after the current limits are applied. The controls for the IBR respond instantly to low voltage caused by the fault and the first two cycles of this response does not have the same current limits as later into the fault. High-speed line relays (i.e. Zone 1 distance or instantaneous overcurrent) can overreach and misoperate during this period. Delaying the relays is one solution to the problem. Another solution is to understand the IBR's transient response to faults to take advantage of the initial current to detect and clear the fault high speed.
- When IBR is connected through a transformer which is a source of zero sequence current, zero sequence ground overcurrent and/or directional protection can be applied as primary ground fault protection. This protection would also take into consideration the mutual coupling effect from the adjacent circuit(s) as in the case of conventional system.
- Depending upon IBR control system design, lack of or non-inductive negative sequence response of IBR poses unacceptable reliability risk where negative sequence overcurrent supervision or directionality elements are used. The negative sequence directional element requires that system unbalance conditions (not corresponding to unbalanced faults) do not jeopardize element security. Some relays need on the order of 10% negative sequence current with respect to positive sequence current to operate reliably. Going forward, transmission owners or regulators can manage the risk by enforcing the IBR suppliers to build their systems to supply negative sequence current proportional to negative sequence voltage unbalance similar to as shown in Figure 8 with line slope setting (k) between 2 to 6 to help alleviate protection element reliability issues. If IBR's are built according to this characteristic emulating apparent negative sequence reactance less than $j0.5$ pu, reliable

operation of negative sequence directional or supervision relay can be achieved with necessary precautions, i.e. by slowing down (if possible) the relay to avoid misoperation during IBR's transient response immediately after the fault.

- Depending on the pick-up settings, use of negative sequence overcurrent protection applications will still pose challenges even when IBR is equipped with the negative sequence current injection control but has low line slope setting, i.e. emulating high apparent negative sequence reactance of about $j0.5$ pu with $k=2$.
- The line current differential is the most reliable protection for a line connecting IBR facility. However, it requires an expensive broadband digital communication channel. It still requires back up by slow local protection – communication independent – for contingency of communication channel failure coincident with the line fault. Backup protection can use phase-to-phase undervoltage and zero sequence overcurrent elements.
- POTT scheme with echo logic and DTT or DCB scheme with DTT from the utility terminal to IBR can overcome reliability risks posed by IBR facility. While both schemes require limited bandwidth, they are not as fast and as reliable as line current differential protection. They require back up by slow local protection – communication independent – for contingency of communication channel failure coincident with the line fault. Backup protection can be phase-to-phase undervoltage and zero sequence overcurrent.
- A DTT from the utility terminal to IBR facility provides fast backup protection disconnecting IBR facility for internal line faults or the breaker failure contingency at the utility terminal.
- Before considering single-phase trip and reclose scheme application on an extra-high voltage line interconnecting a large IBR facility, confirming may be necessary of the IBR's capability to remain connected and supply load unbalance with no risk to the transmission system.
- Use of passive anti-islanding protection based on voltage and frequency measurements often conflicts with voltage or frequency-ride through requirements. Thus, NERC guideline [37] recommends disabling inverter's internal passive anti-islanding protection. If voltage and frequency based anti-islanding protection is not viable, then a DTT based scheme could be installed.
- IBR control system modes can excite subsynchronous resonances in the series compensated lines, particularly when the series capacitor compensated line becomes radially connected to the IBR facility. The system planners to make careful assessment of this risk and call for the custom or special protection needs.
- High IBR penetration can significantly impact system swing characteristics i.e. rate of change of the swing impedance and the swing trajectory.
- Appropriately sized synchronous condensers, operating in parallel with IBR, can overcome system and protection reliability issues. However, initial capital and on-going operational expenses typically limit its application unless coincident with large scale IBR facilities.

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